



Four Seasons Landaa Giraavaru, Maldives
Sustainable Waste Management Plan
Detailed Report

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Written by:	Reviewed by:	Approved by:
Ruben Alvin Joseph	Sripragas Nadaraja	Jessie Tan

1 Executive Summary

The current waste management practices at Four Seasons Landaa Giraavaru is good and segregate & store the generated waste properly. The resort currently disposes most of their solid wastes (except food waste) and liquid wastes (i.e. used cooking and lube oil) to the Thilafushi landfill. Alternatively, items such as polystyrene and cardboard are incinerated on site. The food waste digester is used to convert the food waste into digested liquid effluent and discharge to sewage treatment plant (STP).

The key challenge faced in terms of waste management is the absence of established recycling centres and the challenging logistics involved in transporting wastes from the island to such centres.

On a monthly basis, food, horticultural wastes and used oil/ engine oil account for a significant portion of the mixed waste sent for disposal to the landfill. On average, **27 – 36 tonnes/month** of food and horticultural waste are produced and **1500 L/month** of used oil (i.e., both cooking oil and lube oil) is generated at the resort. The total cost attributable to waste disposal is at least **\$10,200/month**.

To reduce the food and horticultural waste, it is recommended that a food waste composter is procured with a built-in grinder which produces dry solid compost that can be used for landscaping purposes. The cost of recommended 500kg composter is about **\$50,000 USD to \$100,000 USD** with an estimated payback period of between **1.5– 3 years**. Annual cost savings of up to **\$39,000 USD** can be achieved by recycle the food and horticulture water by using composter. This would effectively reduce the amount of horticultural waste sent to the landfill by at least 15 tonnes per month (25% of the total mixed waste sent to the landfill).

To reduce the quantity of used oil disposed, used cooking oil can be sent to a local recycling centre (Secure Bag Pte Ltd) which would provide a total return of \$10.15 USD/ month. Whereas, the quantity of lube oil disposed can be reduced by utilising an oil filter which would extend the oil change interval by 3-4 times. The filter costs is about **\$10,700 USD** inclusive of a 12-month supply of replacement filters. The payback period is **5 months** with an annual savings of **\$25,000 USD**. This would effectively reduce the quantity of lube oil purchased per month by at least **6,300L** (66.5% saving).

The greywater from the laundry water can be utilised by treating to an acceptable level of water quality and then be reused for laundry washing. The proposed system costs about **\$100,000-150,000 USD** and has a payback period of **3-4 years**. An annual savings of **\$49,000 USD** can be achieved with a reduction of **10,000 m³** of freshwater which is equivalent to **8%** of the yearly freshwater generated at the resort.

To complement the solutions proposed above, a suite of in-house management is recommended. These include better segregation of mixed wastes and segregation of recyclable plastics, reducing the incineration of polystyrene and cardboard whilst considering alternatives for polystyrene, reducing single use plastics by utilising locally available natural resources such as bamboo or biodegradable materials such as paper. It is also recommended that the resort enters into contractual agreements with suppliers of electronic products for recovery of used/defective with products.

The consolidated proposal is expected cost an estimated about **\$250,000 USD** with an average payback period of **2-3 years**. Total cost savings per year amounts to an estimated over **\$100,000 USD**. These proposals are expected to reduce the carbon footprint of the resort's activities, in particular its' waste contribution to the landfill. Operational costs are expected to be significantly reduced whilst ensuring a more sustainable waste management plan.

2 Stakeholders

The following **Table 1** summarises the key stakeholders with regards to waste management in Maldives for resorts.

Stakeholder	Role
Ministry of Tourism	Policy maker
Environmental Protection Authority (EPA)	Regulator
Utility Regulatory Body (URA)	Regulator
Baa Atoll Council	Local Administration
Waste Management Corporation (WAMCO)	Waste collector
Parley Maldives	NGO/ Recycler (Plastic)
Secure Bag Pvt. Ltd	Commercial business/ Recycler

Table 1: Key stakeholders for resort waste management in Maldives

Rules and regulations on waste management for resorts in Maldives are gazetted by the Ministry of Tourism (MoT). The regulation that is enforceable to all resorts is The Regulation on the Protection and Conservation of Environment (2006).

The Utility Regulatory Authority Act (URAA) was ratified on the 23 November 2020, under which the Utility Regulatory Authority (URA) has been established as the official authority to regulate and oversee the proper management of public utility services in the country. Services related to waste management will also be governed under the URAA and regulated by URA. As such, resorts in Maldives are bound by the Water Quality for Sewage Disposal Regulations, 1996.

For resorts, waste management practices are bounded by Section 5 of the *Regulation on the Protection and Conservation of Environment in the Tourism Industry (2006)*.

Recently in November 2021, the Ministry of Tourism issued a circular (88-ES/CIR/2021/91) repealing S5.3.3 which allowed the disposal of food waste and biodegradable waste into the sea outside the atoll, considering the wind and ocean currents so that it would not land on the shores of islands. This change is enforceable from 1st January 2022.

Considering the changes as per the circular issued, the following **Sections 2.1 - 2.2** details the existing regulations in place based on waste management, storage of water, sewage, wastewater treatment and the enforceable penalties.

2.1 Waste management

Waste management rules and regulations are found in **Section 5** of the *Regulation on the Protection and Conservation of Environment in the Tourism Industry (2006)* as detailed below.

- 5.1** *Bins to collect waste shall be kept in various areas in an easily accessible manner in all resorts, picnic islands, marinas or such places leased for tourism purposes. Such bins shall be in a clean and sanitary state, with the lid closed.*
- 5.2** *Food and beverages, putrefying items, plastics, paper, glass, iron and items such as cans and toxic or hazardous waste shall be kept in separate bins for each type, and shall be labelled as such.*
- 5.3** *Waste disposal in tourist resorts, picnic islands, and marinas operating in the Maldives shall be carried out in a manner that would have the least impact on the environment, and in accordance with the laws and regulations and in accordance with the following rules prescribed by the Ministry of Tourism and Civil Aviation.*
 - 5.3.1** *Incinerators, compactors and bottle crushers shall be kept and used in all tourist resorts operating in the Maldives. If such equipment does not exist in any resort, such resorts shall obtain and start using that equipment within 6 months from the date of coming into force of this regulation.*
 - 5.3.2** *Waste shall be disposed to the designated area if there is such an area in the region, in the absence of a designated area waste shall be disposed in a manner that is least harmful to the environment.*
 - 5.3.4** *It is prohibited to burn waste generated from the operation of the resort in the open areas of the resort. Incinerator shall be used to dispose such waste. Items that would cause emission of noxious gases into the atmosphere when burned (such as plastics) shall not be burned and shall be collected and delivered to a designated waste management area.*
 - 5.3.5** *Information relating to particulars of vessels, including the capacity and proper logs on trips made for waste disposal in 6 islands or part of it leased for tourism purpose, shall be submitted to Ministry of Tourism and Civil Aviation.*
 - 5.3.6** *It is prohibited to pump any sewer or waste into the lagoons or into any protected area of the ocean from any tourist vessel.*
 - 5.3.7** *Tourist vessels shall dispose sewer and waste in a manner that is harmless to the environment.*
 - 5.3.8** *Every tourist vessel shall have a system for collection and keeping of waste generated in the vessel until such waste is taken to a designated place for deposition of waste. Sewage and wastewater treatment*

Rules and regulations on the storage of water are found in **Section 7** of the *Regulation on the Protection and Conservation of Environment in the Tourism Industry (2006)* as detailed below.

- 7.1** *Sewage shall be disposed in a manner that is least harmful to the environment.*
- 7.2** *In establishing a system for sewage treatment, the following shall be taken into consideration.*

- 7.2.1 Establishing the sewage system in a manner that it would not reach food or living species.*
- 7.2.2 Establishing the sewage system away from the face of the island.*
- 7.2.3 Ensure that no leak is allowed from the sewage system on to the ground of the island.*
- 7.2.4 Ensure that unpleasant smells or other nuisance is not caused.*
- 7.3 A pipe for allowing the smell out shall be kept where septic tanks are used. All septic tanks shall be properly covered.*
- 7.4 There shall be no leak from any junction or sewerage system.*
- 7.5 Sewage water from the toilets shall not be drained into the ground. Treated sewage water shall not be pumped into the lagoon after treatment.*
- 7.6 If sewage water is pumped into the sea, it shall only be carried out at night and shall be pumped out of the reef of the island.*
- 7.7 Sewage water from toilets in tourist vessels shall be deposited into a tank suitable for that vessel, instead of pumping it directly into the sea. The said tank shall be emptied at mid ocean.*

2.2 Penalty

The penalties for the contravention of the said regulation are laid out in **Section 8** of the Regulation on the Protection and Conservation of Environment in the Tourism Industry (2006) as detailed below.

- 8.1 If any provision of this regulation is contravened by any tourist resort, picnic island, marina, hotel, guest house, or tourist vessel, shall be guilty of an offence, and shall be liable to a fine, taking into consideration the seriousness of the non-compliance, between MRF 1000.00 and MRF 10,000.00 in the first instance. Parties repeatedly in non-compliance shall be liable to a fine between MRF 50,000.00 and MRF 100,000.00. If non-compliance of a provision occurs more than once, the Ministry reserves the right to revoke the licence.*

- 8.2 This regulation shall come into force commencing from the 20th day of July 2006.*

2.3 Discharging treated wastewater

Based on the following list of documents, the requirements for infiltration of treated wastewater into aquifers and lagoons are identified.

- Second Schedule of [The Sewage Disposal Regulations 1996](#) by the Ministry of Health and Welfare & Maldives Water and Sanitation Authority
- Annex C of the [General Guidelines for Wastewater Disposal](#) by the Utility Regulatory Authority (URA)

Quality criteria for sewage treatment and disposal systems

Based on the resorts database in 2021, 462 employees were based on FSR Landaa Giraavaru. Thus, including estimates for resort guests and contractors the population size can be conservatively estimated to be between 500 and 1000. The effluent quality requirements are as per **Table 2**.

Note that the effluent criteria are applicable to situations where effluent is disposed into the ground or lagoons with an initial dilution ratio of less than 20:1. Where the initial dilution ratio at the point of discharge exceeds 20:1 the effluent quality requirement shall be as for populations in the range 2000-5000 removal of grease, oil, fats and sedimentation with a maximum concentration of suspended solids not to exceed 50 mg/L.

Population size	Effluent quality requirements
0-1000	Substantial removal of grease, oil, fats. and suspended solids
1000-2000	Removal of grease, oil, fats, and suspended solids. Maximum concentration of suspended solids not to exceed 100 mg/l
2000-5000	Maximum concentrations of BOD (5-day) and suspended solids not to exceed 50 mg/l
>5000	BOD (5-day) concentration of not more than 20mg/l and a suspended solids concentration of not more than 30mg/l

Table 2: Effluent water quality requirements based on population size before disposal based on the Sewage Disposal Regulations 1996

3 Current Waste Management Practices

3.1 Waste management in Maldives

3.1.1 Overview of waste generation

Due to a lack of accurate values per capita on waste generation, the average per capita waste generation is estimated based on mean values from World Bank (2017) and the Ministry of Tourism (2015). These are as follows in **Table 3**.

Type of areas	Waste generated (kg/day/person)	Population in 2020
Urban	2.3	220,000 (40.7%)
Other island communities	0.9	320,541 (59.3%)
Tourist areas (<i>inc. resorts</i>)	3.5	59,947 bed capacity (2022*)

Table 3: Waste generated based on type of area. Sources: [Ministry of Tourism](#), [World Bank](#)

Based on **Table 3**, using an assumed average occupancy rate of 60% throughout a year, the total waste generated per year in 2020 is **336,000 tonnes**, of which 14% is contributed by tourist areas as per **Figure 1**.

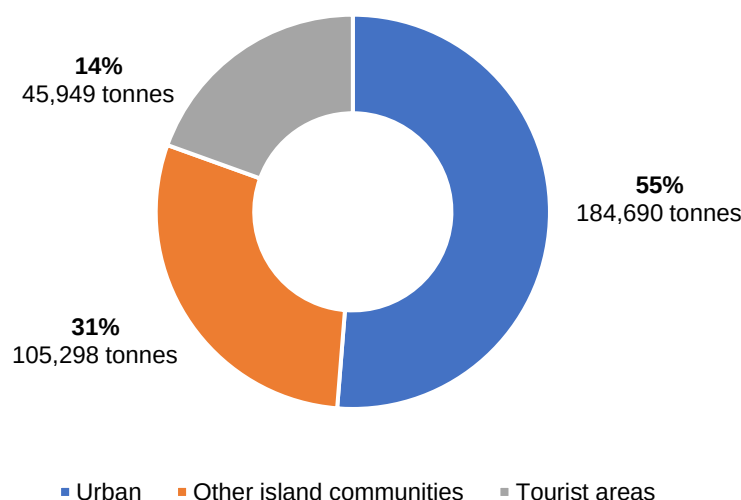


Figure 1: Total waste generated per year in 2020

3.1.2 Waste generation by type of waste for tourism sector

Specific to the tourism industry, compostable general waste such as green waste (vegetable), food waste, landscape waste and horticultural waste are the largest contributor of waste into the landfills at 74%.

This is followed by combustible waste such as paper, cardboard and wood pieces at 9%. The remaining contributors to these are plastic and glass both at 5% respectively.

Other smaller contributors to the waste include mixed waste, metals and e-waste as shown in **Figure 2**.

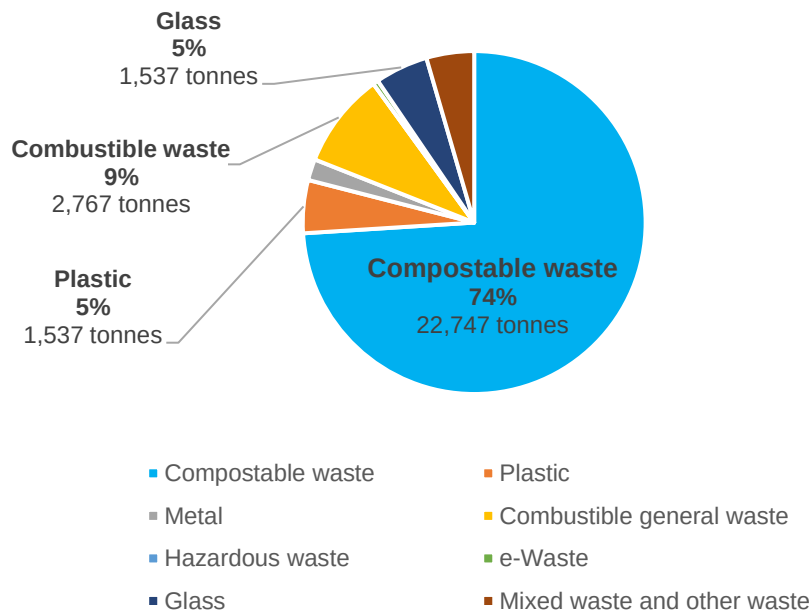


Figure 2: Waste generated according to types of waste by tourism industry in 2018;
Source: [UNESCAP Maldives Waste Record 2018-19](#)

3.1.3 Current waste disposal practices among the tourism industry

a. Before 2022

Prior to 2022, food waste was **allowed to be** disposed into the sea outside the atoll if the wind and ocean currents are considered to ensure the wind and ocean currents would not sweep the waste to the shores of the islands.

Wastes such as horticultural waste (garden & yard trash), construction debris, paper, plastics are usually incinerated or sent to the landfill. Other inorganic waste such as glass, metals, construction debris are mostly sent to the landfill for disposal.

b. 2022 onwards

On the 23rd November 2021, the Ministry of Tourism issued a circular which **prohibited** the disposal of food waste into the sea outside the atoll.

As such, most tourist destinations were required to manage food waste differently. Common methods of managing them are by using food waste digesters or food waste composters. The other types of wastes are disposed in a similar manner as it had been prior to 2022. These are summarised in **Figure 3**.

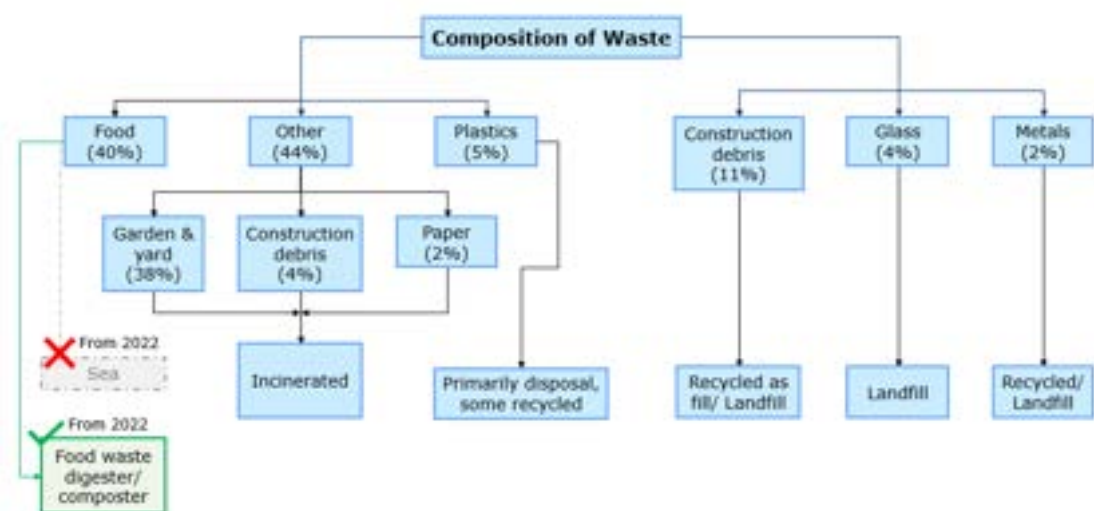


Figure 3: Typical waste disposal methods before 2022; Source: [A Regional Waste Management Strategy and Action Plan for Zone 6 in Maldives](#)

3.2 Current practices at Four Seasons Landaa Giraavaru

3.2.1 Current processes & challenges

Based on the available data from Four Seasons Landaa Giraavaru, the following **Table 4** summarises the average waste generated per month for the year 2021, the disposal methods used and the major challenge faced.

Type of waste	Average amount per month	Disposal	Challenges
Horticultural waste	15 – 22.5 tonnes	Landfill	No national or regional recycling/ reuse facilities of sufficient scale
Food waste	12 – 14 tonnes	Food waste digester	
Kitchen oil	700 Litres	Landfill	
Engine oil (Lube)	800 Litres	Landfill	
Plastics	220kg is recycled (Total quantity unknown)	Recycle/ Landfill	
Metals	Unknown	Landfill	
Glass	Unknown	Landfill	
E-waste	Not significant (Unknown)	Landfill	
Cardboard	Unknown	In-house incinerator	
Polystyrene packaging	Unknown	In-house incinerator	
Activated sludge from STP	90 sacks	Landfill	Generating 7 times more than estimated amount
Laundry water	1,050m ³	STP	Nutrient load causing STP to perform at an unsatisfactory level

Table 4: Current waste generation, processes and challenges faced

Process after segregation of waste

All waste is separated by means of manual separation. After manual separation, certain types of wastes are processed or incinerated using the resort's in-house incinerating facility. They are detailed based on the type of waste as follows with references to **Figure 4 - Figure 11**.

i. Food waste

- Further separation to remove unwanted materials (i.e., bones, plastics and etc.)
- Stored in enclosed storage bins
- Placed inside food waste digester
- Effluent digested food waste is passed to the sewage treatment plant (STP)



Figure 4: Sorting and pre-segregation of food waste

ii. Horticultural waste

- Stored separately on the hotel grounds (i.e., leaves and branches)
- Packed together with mixed waste and sent to landfill



Figure 5: Storage of horticultural waste on the hotel grounds

iii. Used oil

- Used oil (both cooking oil and lube) is stored in a sealed container
- Sealed and sent to the landfill



Figure 6: Storage of used oil

iv. Glass

- Glass bottles and glass utensils for disposal are collected in polystyrene boxes
- Passed to glass crusher to form non-uniform rough edged glass pellets

- Packed in jumbo bags and sent to landfill



Figure 7: Storage and crushing of glassware items

v. E-waste

- E-waste (mainly dry cells) are stored in a container (i.e. polystyrene box)
- Sealed and sent to the landfill



Figure 8: Storage of used dry cells

vi. Polystyrene

- Polystyrene boxes are stored in secluded areas on the hotel grounds
- Incinerated thereafter



Figure 9: Polystyrene boxes collected on hotel grounds

vii. Cardboard

- Stored separately near incinerator
- Incinerated thereafter (together with polystyrene boxes)



Figure 10: Cardboard boxes stored nearby incinerator

viii. Mixed waste

- Typical proportion includes wet waste (50-60%), horticultural waste (40%), other waste such as construction waste, debris, discarded items, dust from incinerator are stored in a secluded area on the hotel grounds (<5%)
- Packed in jumbo bags
- Sent to landfill



Figure 11: Mixed waste temporarily stored

Disposal to landfill

Figure 12 illustrates the summary of waste disposal to the landfill across the months in 2021. Waste was highest in the month of December due to typical seasonal uptick in demand. Maldives. In the months of May and June, high waste was generated due to reported construction works that had taken place and horticultural wastes despite reduced occupancy because of the outbreak of newer variants of COVID-19 virus across the globe. On average, the waste generated per month amounts to 60 tonnes.

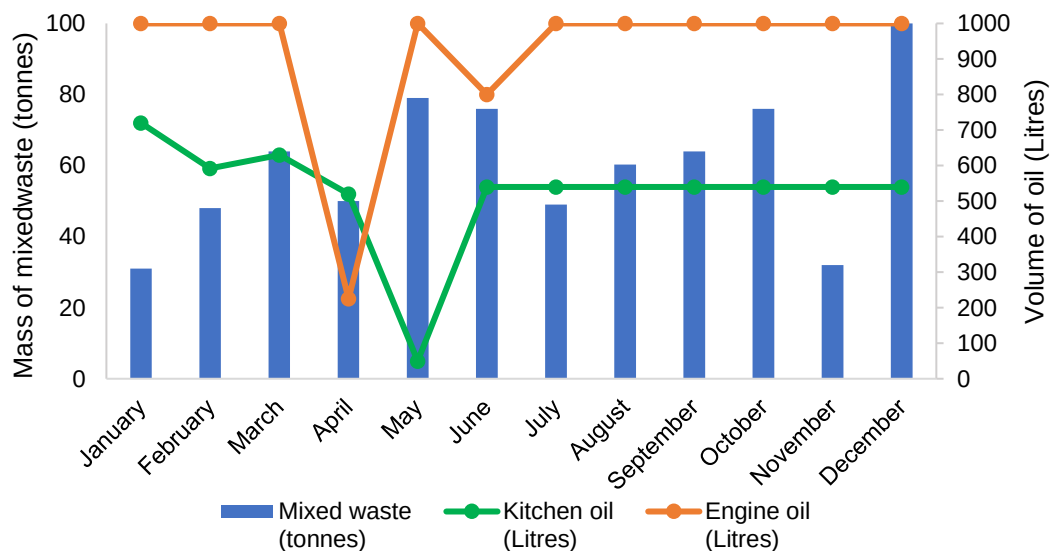


Figure 12: Waste disposal at FS-LG by month in 2021

3.2.2 Quantity & costs

The disposal cost of waste to the landfill includes a fixed cost and a variable cost. These are as follows in **Table 5**.

Type of cost	Payee	Cost (\$USD)	Details
Fixed	WAMCO	2,290/month (\$286.25 USD per trip)	Gate fee into Thilafushi
Fixed	Garbage “dhoni” (Boat service)	4,030/month (8-trips per month; each trip carries 3-4 tonnes of waste)	Fixed cost regardless of number of trips; used also for transporting goods
Variable cost	WAMCO	Depending on tonnage	Not same as \$286.25 USD, depending on WAMCO rates for boat
Variable cost	Garbage “dhoni” (Boat service)	1,300/additional trip	N/A

Table 5: Cost structure for the current waste disposal arrangements at FS-LG

Since at least 2-3 additional boats are also hired on an ad-hoc basis each month, the total estimated cost for waste disposal is \$10,200 USD per month.

4 Proposed solutions

4.1 Food waste

The summary of the intermediate solution for the digested food waste effluent has been detailed in **Annex A**.

For the permanent solution, the reader is advised to refer to **Section 4.4** which details the use and costs of a food and horticultural waste composter.

4.2 Used cooking oil

Due to the nature of the tourism industry and brand name of Four Seasons, reusing recycled (filtered & purified) cooking oil may not be in the best interests of the hotel. This is because, food cooked with recycled oil tends to have a lower palatability than that of pure cooking oils.

With Four Seasons being a leader in the chain of luxury hotels, the usage of recycled oil in the preparation of foods may adversely affect guests' experience with respect to the food and beverages served within the premises of the hotel.

a. Option 1 – SBPL

Used cooking oil can be sent to **Secure Bag Pvt Ltd (SBPL)**, a local recycling collection centre located in Male, Maldives. This option can be pursued imminently.

Process

- i. An agreement between Four Seasons and SBPL needs to be made prior to any sending of acceptable recyclables.
- ii. Recyclables are to be sent to Thilafushi as it had been previously.
- iii. Recyclables are collected by SBPL personnel and are to be weighed and recorded accordingly (collected and sent to India for further processing).
- iv. Receipts are sent to the resort.
- v. Monetary compensation for receipts is directly sent to the company accounts.

Estimated return

A drum of used cooking oil is valued at MVR 45/per 200L (\$2.9 USD/200L) by SBPL. The average amount of used cooking oil generated by the hotel is approximately 700L, thus the amount that would be received on average would be **\$10.15 USD/month**. Information on the list of recyclables and their equivalent rates are listed in **Figure C 15**.

4.3 Used lube oil

a. Option 1 – GLW Global

Used lube oil can also be filtered and reused for the generators on the hotel grounds. There are currently 5 gensets in Four Seasons Landaa Giraavaru which requires approximately 800L of lube per month. Each generator can be used for an estimated 250 hours before an oil change is required. This is better known as the oil change interval period.

With the use of the filter below in **Figure 13** the generator can be used up to 3-4 times longer before an oil change is required. The process and costs of the filter are detailed below.



Figure 13: Oil filter from GLW Global; Source: [GLW Global](#)

Process

- i. Lube from the oil sump is flowed into a filter which traps particles larger than 1 micron in size.
- ii. Water is evaporated through the vents and the oil exits and returns to the oil sump.
- iii. The oil can be reused once again.
- iv. GLW filter and existing OEM filters need to be replaced once in every 750-1,000 hours of use.

Proposed usage plan

On the hotel grounds, there are 5 gensets at present which are tabulated as follows in **Table 6**.

Genset #	1	2	3	4	5
Model	KTA38-G7	KTA50-G8	KTA38-G3	KTA19-G4	KTA38-G5
Oil change interval (hours)	250	250	250	250	250
Average genset running hours (hours/month)	375	375	375	270	400
Engine oil sump size (Litres)	135	177	140	50	140
Quantity of OEM lube oil filters	6	7	6	4	6
Total cost for OEM filter per oil change (\$ USD)	62	92	62	52	62

Table 6: Gensets on hotel grounds

With the use of a GLW oil filter, the oil change interval can be extended by up to 3-4 times (in hours). As such, oil change will only be required after 750-1000 hours. The cost of the GLW filter and the payback period are highlighted below in **Table 7**.

Capex (\$ USD)	10,700 (Inclusive of 1-year supply replacement filters)
Maintenance cost (\$ USD)	6,100/year (GLW filter consumables purchases per year)
Payback period (Months)	5
Average cost savings (\$ USD)	~27,000/year
Reduction in lube oil purchase per year	16,760L (84%) Equivalent of \$63,700 USD of lube

Table 7: Cost of oil filter and payback period

The key assumptions and parameters used in determining the payback period are as follows:

- Cost of oil disposal to the landfill is \$0.3 USD/Litre
- Genset is modelled to be functioning in continuous hours (i.e., no breakdowns) by extrapolating running hours into monthly period for ease of calculation (*Genset currently functions at an equivalent rate of 15 days/month while oil change interval is 11 days/month*)
- Running time of gensets is taken to be the average of all 5 gensets (i.e., ~360 hours)
- Both OEM filters and GLW filters will be required to be changed upon exceedance of the oil change interval

The key changes in the function of the gensets and the oil change requirements are highlighted as per **Table 8**.

	Current system (OEM filters only)	Proposed system (OEM filters & GLW filters)
Genset operating period per month (hours)	360	
Genset operating period per month (equivalent days)	15	
Oil change interval (hours)	250-300	750-1,000
Oil change interval (days)	11-13 (approx.)	31-42 (approx.)
Lube oil required (Litres/month)	800 (based on average monthly requirements)	270 (approx.)
Average cost of filter replacement (\$ USD/month)	330 (OEM filter changed once per month only)	1,347 (OEM & GLW filters changed once in every 2 months)

Table 8: Comparison of operating parameters before and after implementation of GLW filters

b. Option 2 – SBPL

Used lube can also be sent to **Secure Bag Pvt Ltd (SBPL)**, a local recycling collection centre located in Male, Maldives.

Process

- An agreement between Four Seasons and SBPL needs to be made prior to any sending of acceptable recyclables.
- Recyclables are to be sent to Thilafushi as it had been done previously.
- Recyclables are collected by SBPL personnel and are to be weighed and recorded accordingly (collected and sent to India for further processing).
- Receipts are sent to the resort.
- Monetary compensation for receipts is directly sent to the company accounts.

Estimated return

A drum of used engine oil/diesel is valued at MVR 45/per 200L (\$2.9 USD/200L) by SBPL. The average amount of used cooking oil generated by the hotel is approximately 800L, thus the amount that would be received on average would be **\$11.6 USD/month**.

4.4 Food and horticultural waste

To date, 12 different composter vendors have been engaged for a suitable solution to manage the food and horticultural waste in FSR Landaa Giraavaru.

This assessment was conducted based on the composter being able to convert food and horticultural waste into fine and dry compost with minimal to no liquid effluent, odour and etc.

From these 12 products, we have shortlisted 4 vendors for further assessments and interviews. These vendors include ZWaste, Westcom, ECOBOT and MAECO.

Based on our assessment, composters offered by ECOBOT and MAECO are recommended.

It is strongly recommended that the final decision is supplemented by the feedback received from site visits to both the vendor's past or existing projects respectively.

The reader is advised to refer to the attached Appendix A – Food waste composter study for FSR LG.

The payback period and annual savings for composter from ECOBOT and MAECO are as follows below in **Table 9**; with reference to Annex B.

	Payback period	Annual savings
ECOBOT	1 year 4 months	\$39,000 USD
MAECO	2 years 4 months	\$37,000 USD

Table 9: Payback period and annual savings for ECOBOT and MAECO composters

4.5 Laundry water

There are two available options that can be used to treat the laundry water. The first option is to reuse the treated water for laundry washing. The second option is to use a primary treatment system which would act as a pre-filter before passing the treated water to the STP.

a. Option 1 – Reuse

The greywater from the laundry can be treated to be reused for laundry once again as per **Figure 14**.

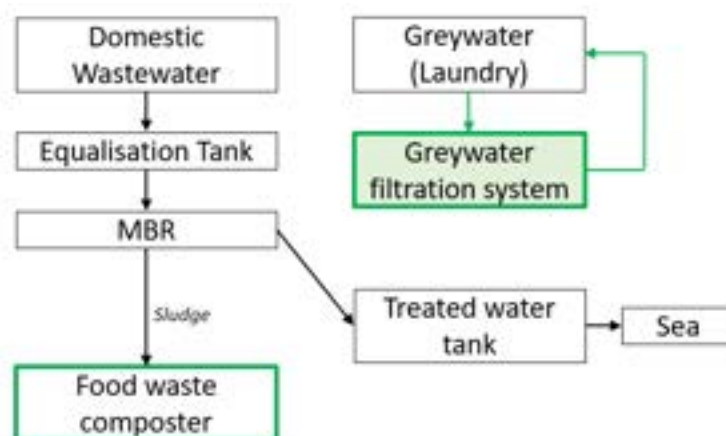


Figure 14: Process flow of laundry water from the resort (Option 1)

Process

- i. The laundry water is subject to chemical treatment to adjust the pH to prepare for coagulation and flocculation where small particles are clumped for easier removal.
- ii. Solid and liquid are separated through the dissolved air flotation mechanism (DAF).
- iii. Particles of sizes between 0.01 – 1,000 μm are filtered.
- iv. Water is dechlorinated to remove foul odour and turbidity through the activated carbon filter.
- v. Further disinfection is carried out to remove pathogens through advanced oxidation.

Proposed usage plan for the greywater treatment system

The resort currently generates an estimated 35m³/day of greywater while the capacity of the system is only 50m³/day. Thus, all the water used from laundry is to be treated by the system and be reused.

A detailed assessment of the effluent laundry water quality needs to be carried out and be provided to the equipment supplier to ensure the system can treat it (or remedial design solutions would be proposed). Such an assessment should encompass water quality parameters such as BOD, COD, TSS, TDS, oil and grease, pH and temperature.

Costing and pricing

The greywater system's specification and costs are detailed as per **Table 11**. It is to be noted that the cost quoted is inclusive of the equipment, installation, commissioning supervision, training and 12-month warranty.

The current quote requires the resort to be responsible for the following scope of works:

- Site clearance
- Build ancillary infrastructure (Tanks for raw water and treated water tank, plinth for system, shelter for system)
- Piping and electricity supply works
- Others including manpower, logistics support from port to site, approvals from regulators, maintenance, operations etc.)

Capacity (m³/day)	50
Actual flow rate (m³/day)	35
Floor area required (m²)	50
Capex (\$ USD)*	100,000-150,000
Maintenance cost (\$ USD)	3,600
Average cost savings (\$ USD)	49,000/yearly
Daily power requirements (kWh/m³)	1.2
Payback period	3 year 10 months
Quantity of water saved (m³)	9,950m ³ / year

Table 10: Proposed usage plan for the greywater treatment plant

The key assumptions and parameters used in determining the payback period are as follows:

- Cost of freshwater generation is \$2.6/m³ (Estimated based on data provided by the resort)
- Cost of electricity is assumed to be \$0.2USD/kWh (~\$756/month to power the treatment system per month)
- Practical water recovery rate is taken to be 85% of incoming greywater (Note: only 2% is typically lost according to previous tests. A conservative loss of 15% is taken)

b. Option 2 – Pre-filter

The greywater from the laundry can be pre-treated and then be passed into the MBR STP as per the following **Figure 15**.

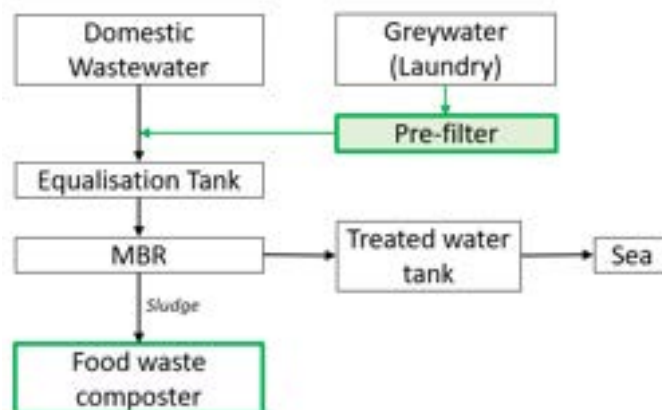


Figure 15: Process flow of laundry water from the resort (Option 2)

Process

- i. Effluent water is passed through bar screen chamber to remove large debris.
- ii. Filtrate flows into collection cum equalization tank.
- iii. The water is pumped to reaction tank and chemicals are added with mechanical agitation.
- iv. The reacted effluent will be overflowed to dissolved air flotation (DAF) system for sedimentation and oil removal.
- v. The resulting clear water flows to bioreactor for secondary treatment; settled sludge is discharged to sludge dewatering system.

Proposed usage plan for the greywater treatment system

The resort currently generates an estimated 35m³/day of greywater while the capacity of the system is only 30m³/day. The balance 5 m³/day is to be disposed into the STP as it has been previously.

A detailed assessment of the effluent laundry water quality needs to be carried out and be provided to the equipment supplier to ensure the system can treat it (or remedial design solutions would be proposed). Such an assessment should encompass water quality parameters such as BOD, COD, TSS, TDS, oil and grease, pH and temperature.

Costing and pricing

The greywater system's specification and costs are detailed as per **Table 11**. It is to be noted that the cost quoted is inclusive of the equipment, installation, commissioning superviseon, training and 12-month warranty.

The current quote requires the resort to be responsible for the following scope of works:

- Site clearance
- Disposal of treated water
- Provision of freshwater
- Piping and electricity supply works
- Others including manpower, logistics support from port to site, approvals from regulators, maintenance, operations etc.)

Design flow rate (m³/day)	30
Actual flow rate (m³/day)	35
Floor area required (m²)	140
Capex (\$ USD)	100,000-150,000
Maintenance cost (\$ USD)	3,000 (estimated cost of consumables)
Average cost savings (\$ USD)	No cost savings as no reduction of freshwater use
Daily power requirements (kWh)	33
Payback period	N/A

Table 11: Proposed usage plan for the greywater treatment plant

c. Recommendation

Option 1 and option 2 are two different systems as option 1 allows for reuse and option 2 acts as a pre-filter. Thus, a like-for-like comparison is not feasible.

However, based on **Table 12**, it is proposed that option 1 is pursued over that of option 2. This is because for a marginal cost different, annual savings up to \$49,000 can be achieved whilst also reducing the demand for freshwater from the desalination plants in the resort. Besides, the investment has a payback period of 3-4 years and will improve the resorts water management resiliency.

	Option 1 (Reuse)	Option 2 (Pre-filter)
Capacity/ Design flow rate (m ³ /day)	50	30
Maintenance	Requires weekly/ monthly checks on equipment and consumables condition	Requires weekly/ monthly checks on equipment and consumables condition
Capex Cost (\$ USD)	100,000-150,000	100,000-150,000
Yearly maintenance (\$ USD)	3,600	3,000
Lifespan	20 years	
Total cost over 10-year period (\$ USD)	241,000	205,000
Cost per volume over 10- year period (\$ USD/m ³)	0.66	0.94
Monthly savings (\$ USD)	3,800	N/A
Annual savings (\$ USD)	49,000	N/A
Payback period	3-4 years	N/A
Reduction in freshwater demand (m ³)	~10,000 m ³ / year	N/A

Table 12: Comparison of greywater treatment systems

4.6 Plastics

a. Reducing storage space of plastics

To reduce storage space of plastic waste, it needs to be properly segregated at source to ensure a minimal quantity is part of the mixed waste sent to the Thillafushi landfill.

The plastic waste is recommended to be segregated into 2 categories as follows:

- **Recyclable plastic**
 - *PE*: Polyethylene Terephthalate & *HDPE*: High density polyethylene (Examples of *PE*: Shampoo bottles, mineral water | Examples of *HDPE*: Detergent bottles, plastic bags, chemical bottles)
- **Non-recyclable plastic**
 - *LDPE*: Low density polypropene & *PP*: Polypropene (Example of *LDPE* and *PP*: Domestic plastic bags)

The first category (recyclable plastics) can be easily processed for recycling whereas the second category cannot be easily processed for recycling. An example of these plastic categories are illustrated in **Figure 16**.



Figure 16: Classification of plastic types

The accumulated plastics are recommended to be passed into a plastic shredder to reduce the volume of the resulting plastic mass. The shredded plastic waste can then be stored in barrels and stored at the back of house of the resort. This would significantly reduce the space required to store the plastic wastes.

b. Continuing recycling with Parley

The resort has an existing agreement with an international NGO, Parley where all recyclable plastics are sent to be reused or recycled every month. For the year 2021, an estimated 2.6 tonnes of plastics from Four Seasons Landaa Giraavaru were sent to Parley (~210kg/month).

It is recommended that the resort continues this arrangement with Parley and introduce several in-house policies as per **Section 4.8** to further reduce the plastic wastes generated.

4.7 Metals, glass & e-waste

Metals, glass and e-waste (i.e., batteries) can also be sold to SBPL based on the following process with their respective rates as listed below:

Process

- i. An agreement between Four Seasons and SBPL needs to be made prior to any sending of acceptable recyclables.
- ii. Recyclables are to be sent to Thilafushi as it had been previously.
- iii. Recyclables are collected by SBPL personnel and are to be weighed and recorded accordingly (collected and sent to India for further processing).
- iv. Receipts are sent to the resort.
- v. Monetary compensation for receipts is directly sent to the company accounts.

Estimated return

The rates offered by SBPL for each of the following materials are as follows:

- Metals: 2.5MVR/kg (\$0.16USD/kg)
- Glass: 0.25MVR/kg (\$0.016USD/kg)
- E-waste: 7MVR/kg (\$0.45USD/kg)

4.8 Holistic in-house waste management policies

On top of the proposed solutions detailed above, it is also proposed that the resort develops several in-house policies to reduce wastes such as plastics and e-waste as listed below and ensure its current disposal and segregation processes pose the least harm to the environment.

Segregation of mixed waste

As detailed in **Section 3.2.1**, the mixed waste can be better separated to ensure less waste is sent to the landfill and more waste can be recycled.

The horticultural waste can be further separated and passed into the food waste composter. This would remove 40% of the bulk quantity within the mixed waste sent to the landfill.

Debris and discarded items such as compressors, metal sheets can be unassembled and sent to SBPL for recycling.

This would effectively reduce the quantity of the mixed waste sent for disposal to the landfill by an estimated 50% than the typical quantities.

Incineration of polystyrene

The adverse effects of polystyrene incineration have been well documented to release harmful emissions including polycyclic aromatic hydrocarbons (PAHs).

It is recommended that Four Seasons Landaa Giraavaru considers the following options to reduce the need to incinerate polystyrene/ styrofoam boxes.

- Return polystyrene boxes back to suppliers
- Consider utilising polystyrene alternatives such as mycelium packaging (a mushroom based biodegradable product), perforated cardboard or recycled shredded paper as per **Figure 17**



Mycelium packaging



Perforated cardboard



Recycled shredded paper

Figure 17: Alternatives to polystyrene packaging

Cardboard

At present, there are no companies which recycle paper or cardboard. As such, it is recommended that the usage of cardboards is reduced by ensuring cardboard boxes are returned to suppliers respectively where possible.

E-waste

Besides sending e-wastes to be recycled through SBPL, it is also proposed that the resort enters into contractual agreements with suppliers of electronic devices for recovery schemes (i.e., television sets, walkie talkies, laptops and etc) where a machine/ equipment/ device is returned to the supplier for disposal/ recycling once it is no longer in usable condition.

Plastic

The resort has an agreement with Parley to send recyclable plastic waste to them. To further reduce plastic wastes, it is recommended that the following policies are implemented.

- Stop single use plastics; replace with naturally sourced items which can be biodegradable (i.e., bamboo etc.) as per **Figure 18**.
- Use compostable cutlery and food packaging as per **Figure 18**.



Bamboo-based vanity kit



Paper-based food packaging

Figure 18: Options to reduce plastic usage

4.9 Consolidated costs and savings of recommended solutions

To illustrate the financial viability of the proposal for food & horticultural waste, lube oil recycling and laundry water recycling.

Table 13 illustrates that the total cost required to procure the relevant machine/ equipment amounts to a total of \$220,000USD. The cost payable on a yearly basis for maintenance and consumables is approximately \$10,000 USD. The average savings that can be made in one year is approximately \$116,000USD. As such, the consolidated payback period for the 3 solutions is **2 years**.

Beyond the payback period, at least \$116,000 USD worth of operational costs will be reduced per year. Moreover, a total of at least 15 tonnes/month (25% of total monthly mixed waste sent to landfill) of mixed waste and 530 litres/ month (47% of total used oil per month) of used oil will not be sent to the landfill monthly. The demand for freshwater will also reduce by 10,000 m³/year (8% of the freshwater demand per year) which greatly reduces the load and reliance on the desalination plant.

	Capex cost (USD)	Yearly maintenance cost (USD)	Avg. savings/benefit per month (USD)	Average savings/benefit per yearly (USD)	Payback period	Reduction in waste/ resources used
500 kg composter (ref.ECOBOT)	\$46,643	\$185	\$3,006	\$39,075	1 year 4 months	15 tonnes/ month of food & horticultural waste (25% of mixed waste)
Lube oil filter (ref. GLW)	\$4,565	\$6,106	\$2,100	\$27,914	5 months	530 Litres/month of lube oil (35% of used oil)
Laundry water recycling (ref. ECOSOFT)	\$169,000	\$3,600	\$3,800	\$49,200	3 years 9 months	Reduction ~10,000 m ³ /year of freshwater (8% of yearly freshwater demand)
Total	\$220,208	\$9,891	\$8,906	\$116,189	2 years	

Table 13: Consolidated costs and savings for the proposed solutions for the resort

5 Conclusion

The current waste management practices at Four Seasons Landaa Giraavaru is good and segregate & store the generated waste properly. The resort can adopt a more sustainable approach to their waste management by implementing a suite of solutions which encompass the installation of equipment/devices and the implementation of in-house management policies.

The largest contributor of waste in the resort is the food and horticultural waste. A significant proportion of the waste sent to the landfill can be reduced using a food & horticultural waste composter. This reduces the monthly costs of waste disposal and more importantly reduces the burden on the Thilafushi landfill.

Used oil also contributes to a sizeable proportion of the waste sent to the landfill. The use of filters will extend the lube oil change interval thereby reducing the quantity of required per month. This will reduce the outgoing waste from the resort and reduce the operational costs with regards to the replacement and purchase of lube oil. On the other hand, cooking oil can be sold to a local recycling collector – Securebag Pvt Ltd at a premium.

The laundry water can be recycled with the use of the appropriate treatment system to be reused for laundry washing. This will reduce the load of STP and RO treatment plant by 8%.

Items such as metals, glass and e-waste can also be recycled through Secure Bag Pvt Ltd. Plastic items should continue to be recycled through the existing partnership with Parley.

To complement the technologies' and equipment proposed it is also recommended that the resort implements a suite of in-house management policies to assist them in further reducing the waste generated. These encompass better segregation of mixed waste, stopping the incineration of polystyrene and cardboard, entering recovery agreements with suppliers of electronic & electrical devices, reducing single use plastics by utilising locally available materials such as bamboo or biodegradable materials such as paper.

It is hoped that the proposed solutions be given its due consideration to ensure the waste management plans in Landaa Giraavaru is sustainable.

There are two other key areas, sustainable water management and energy management are recommended to study in detail to implement / enhance the sustainable solution / operational practices respectively at Four Seasons Landaa Giraavaru to reduce the operational utility cost and carbon emission to the environment.

Annex A

Sustainable Waste Management Plan Intermediate Report – Wastewater



Four Season Landaa Giraavaru, Maldives
Sustainable Waste Management Plan
Intermediate Report - Wastewater

Ref: Web(E)/108/01/WMP/GG

Date: April 2022

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Written by:	Reviewed by:	Approved by:
Ruben	Sripragash	Jessie Tan

1 Executive Summary

Four Seasons Landaa Giraavaru employs an in-house sewage treatment plant (STP) using a membrane bioreactor (MBR). At the point of reporting, domestic wastewater (*blackwater*), laundry water, and aerobically digested food waste effluent are passed into the STP.

The key issues faced include i) strong unpleasant odour, ii) excessive foaming, iii) dewatering of solids. To address these issues, Web Earth has developed several proposals incorporating sustainable elements into the existing waste management plan.

The odour issue has been reported since day 1 but the strong unpleasant odour reported after the food waste effluent discharge into the STP. The odour is generated from gasses from anaerobic reaction at equalization tank and high organic loading due to the digested food waste liquid effluent.

With regards to the unpleasant odour, an intermediate solution and a permanent solution is proposed. The intermediate solution entails the use of an oxidizing agent (*LIMIN8/ liquid Potassium Permanganate*) in the equalization tank and allow oxidized gasses (no odour) to be released through a vent pipe. Further to that, 4 openings along the equalization tanks are to be made to provide aeration into the tank using the existing air pump for the MBR.

The permanent solution to the odour from the STP would be to utilise the openings used; whereby 2 of them are used for aeration and the remaining 2 openings are inserted with vent pipes.

The permanent solution for the digested liquid effluent from the food waste would be to utilise a constructed wetland (with a plan area of 28m²). This wetland will receive the effluent from food waste digester which typically amounts to 4-5m³/day. Suitable plant species will be planted to allow nutrients from the food waste effluent to be absorbed. The effluent will then be allowed to infiltrate into the ground via the soakaway.

To reduce the excessive formation of foams, two options are proposed. The first is to pass the laundry water into a pre-filter and thereafter allowing it to flow into the STP – MBR. The second option would be to filter the water more thoroughly and in a separate mechanism. This treated water can then be used for non-potable water purposes.

To ensure longer term sustainability, it is proposed that Four Seasons Landaa Giraavaru considers the adoption of a food waste composter to completely digest the food waste and thereafter be used as fertiliser or for landscaping purposes. Alongside the composter, it is recommended that an ultrafiltration system be adopted to allow the laundry water to be reused for non-potable uses such as toilet flushing and other appropriate uses.

It is proposed that these solutions be adopted to ensure the pressing issues surrounding the wastewater management system in Four Seasons Landaa Giraavaru are addressed soon. Adoption of these measures would also steer the resorts operations towards a more sustainable and circular system.

2 Current Situation

2.1 In-situ conditions and infrastructure

Four Seasons Landaa Giraavaru currently utilises a membrane bioreactor (MBR) as part of the sewage treatment plant. Along with the MBR, an aerobic food waste digester (FWD) from Power Knotts is being used to digest the food waste generated from the resort. The following **Table 1** highlights some of the key parameters of the current system.

	STP (Membrane bioreactor)	Food waste digester (FWD)
Year of installation	2018	2021
Capacity	400m ³ /day	500kg/day
Input	- Domestic wastewater (<i>blackwater</i>) - Laundry water (25 – 30 m³/day) - Food waste effluent (5m³/day)	400 – 800kg
Output as per product specifications	3bags/ week	2m ³ /day
Actual output	3bags/ day (Before FWD)	4m ³ /day
Additional information	N/A	Requires 3 pax manpower per day for manual separation

Table 1: Key parameters of the MBR and FWD

The domestic wastewater, laundry water, and food waste digester effluent are being passed to directly to the STP. First, it is passed to the equalization tank where it acts as a buffer by collecting the widely fluctuating incoming raw effluent and supplies a steady flow rate into the STP.

To date, Four Seasons Landaa Giraavaru faces 3 main problems with the STP. These include the following:

1. Strong unpleasant odour which has worsened since FWD effluent was passed into the STP
2. Excessive foaming
3. Dewatering of solids (i.e., difficulty of removing solids/ sludge)

The following sections of the report dissects the issue further along with proposals to solve these issues.

2.2 Current wastewater management system on site

Following the installation of a food waste biodigester, three different streams of wastewater are fed into the equalisation tank – food waste digester effluent, domestic wastewater and wastewater from laundry as per **Figure 1**.

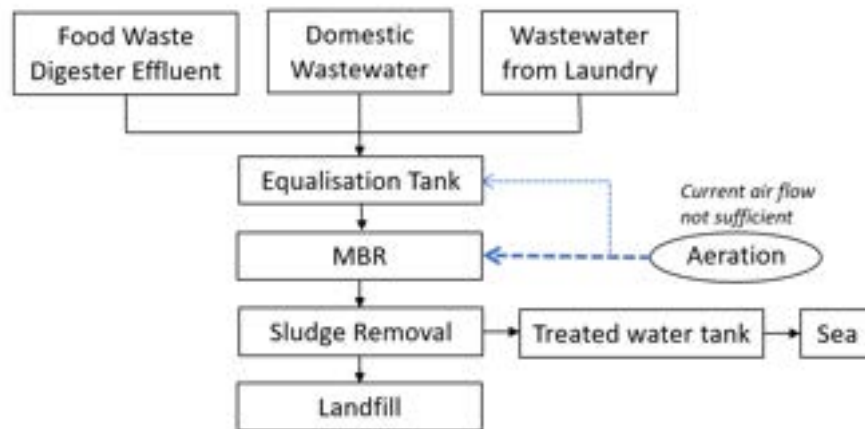


Figure 1: Current wastewater management system

The equalization tank is fully enclosed, and the wastewater is passed to the membrane bioreactor (MBR). The sludge is removed and sent to the landfill and the effluent treated water is stored in a tank and is released to the sea thereafter.

Odour issue

The addition of the food waste digester effluent has caused the level of organic load into the sewage treatment to increase. As such, a pungent-smelling odour has been observed on site.

The odour generated is due to an increase in the organic load and the limited aeration into the equalization tank. As such, upon exiting the equalization tank and prior to flowing into the membrane bioreactor (MBR), these odours are released into the air causing great discomfort to guests and workers from the resort.

Foaming issue

Besides the odour, appreciable foams can be observed on the surface of the wastewater at MBR. This is likely to be due to the presence of suds-forming detergents (caused by filamentous bacteria growth) and soaps within the influent laundry water. Thus, reducing the effectiveness of the treatment from the MBR. To address the excessive foaming, staff on-site utilise an anti-foaming agent (NALCO 71D5 PLUS) to reduce the excessive foams produced.

Solid dewatering issue

The sludge from the STP is also becoming increasingly difficult to separate from the liquid wastewater. It has recently been observed that more solids are now part of the STP system, possibly due to the introduction of the food waste digester effluent and the laundry water.

3 Proposed solutions

3.1 Intermediate solution proposal for odour issue

As a temporary measure, Web Earth recommends to cut the supply of food waste digester effluent into the STP as per **Figure 2** (a separate solution to treat the food waste digester effluent has been detailed in **Section 3.3**). This will immediately reduce the organic load into the STP.



Figure 2: Temporary wastewater management system

Aeration is to be provided into the equalization tank as per the steps detailed below and based on **Figure 3**. The detailed method statement for the works of this intermediate solution has been detailed in **Section 4.1**.

- Access opening of the tank to be closed
- A pre-determined dosage of LIMIN8/ liquid Potassium Permanganate (an oxidizing agent in liquid state) is to be added
- STP is to be left for a couple of hours
- Openings of the appropriate diameter are to be cored from the concrete cover overlaying the equalization tank
- To utilize the spare capacity of air being currently supplied into MBR

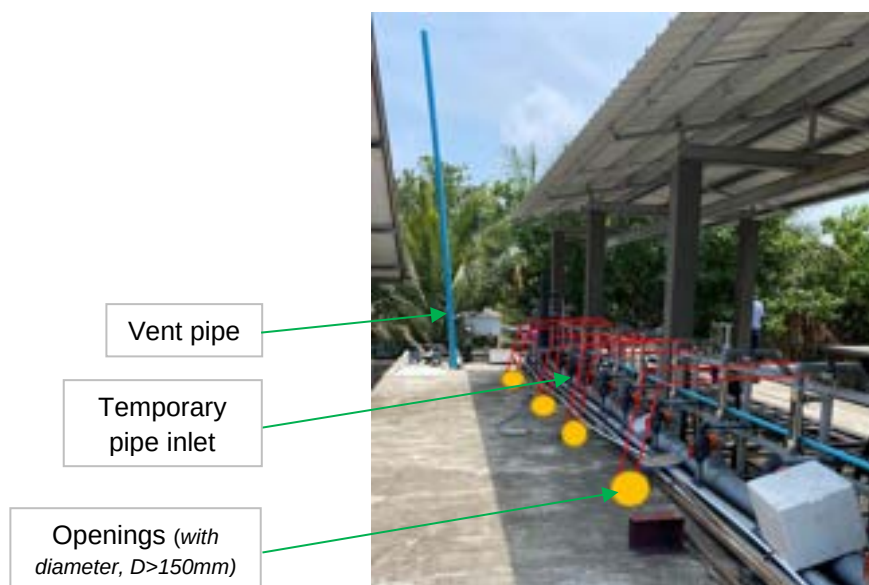


Figure 3: Demarcation of openings and vent pipe

3.2 Future and permanent solution proposal for odour issue

Upon the successful execution of the temporary solution detailed in **Section 3.1**, it is recommended that a more permanent solution is further explored.

The permanent solution entails the provision of adequate vent pipes and installing an air blower directly into the equalization tank as shown in **Figure 4**.

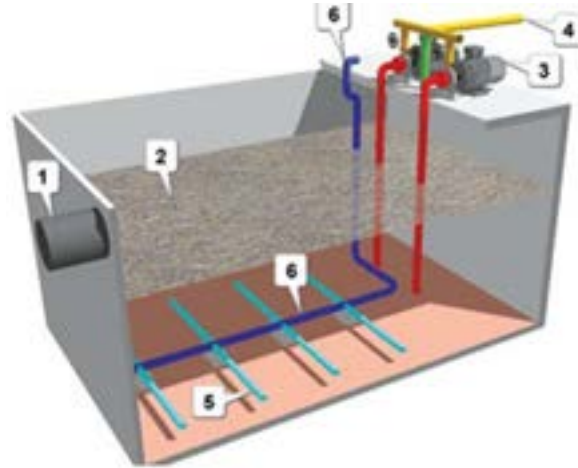


Figure 4: Air blower in the equalization tank

The openings made as detailed in **Section 3.1** can be used to insert the air directly into the bed of the equalization tank. The existing gas pump can be continued used if its capacity is sufficient for the aeration required for the tank. However, should the capacity be insufficient to meet the demand for aeration required for both the MBR and the equalization tank, a pump of sufficient capacity is to be procured and installed accordingly.

3.3 Constructed wetlands for food waste digester effluent

The food waste digester effluent is proposed to be diverted to a treatment train which begins with a grease trap, followed by a constructed wetland and a soakaway to allow for ground infiltration as per **Figure 5**.

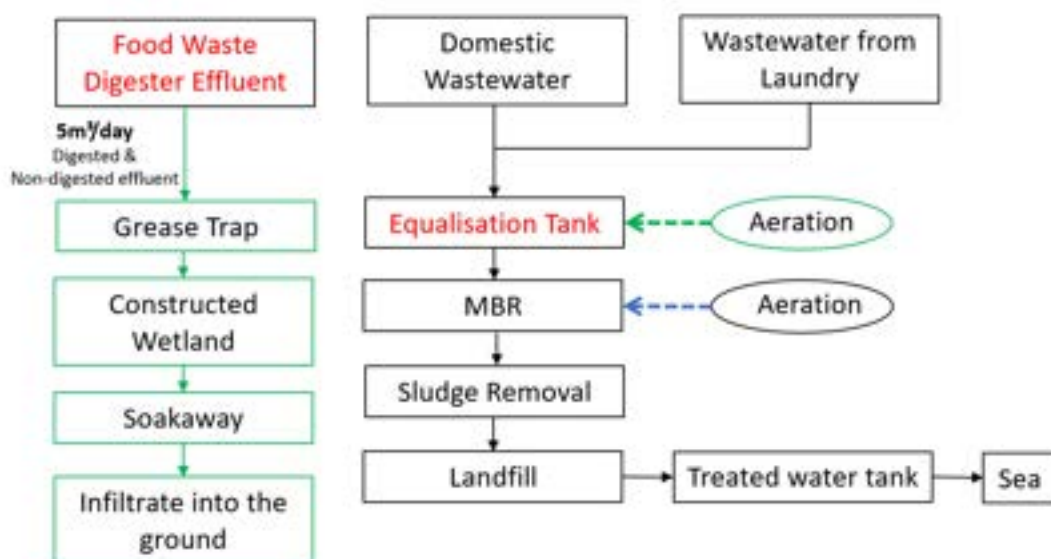


Figure 5: Treatment train for food waste digester effluent

The proposed constructed wetland (CW) is a subsurface flow wetland covered with topsoil to ensure any odour to the surrounding areas is greatly minimized. The details of the constructed wetlands are as attached in **Annex A – Proposed Constructed Wetland**.

The key design parameters of the constructed wetland are summarised as follows:

- Dimensions: Length = 7m; Width = 3.5m; Depth = 0.6m
- Perforated inlet pipe: Pipe diameter = 100 mm; Number of perforations = 186, Slot opening (width = 3mm, length = 7.5mm), Openings per meter = 4185mm²
- Detention time in CW: >24 hours
- Side slope: 1:2
- Base slope: 1:100
- Sump dimensions: Length = 0.8m; Width = 0.8m; Depth = 0.5m
- Soakaway dimensions: Length = 2m; Width = 3.5m; Depth = 1m (as per **Figure 7**)
- Liners required: 12mm plastic liner for CW; Geotextile for soakaway
- Plant species: *Vetiver zizanioides* or *Canna indica* or *Phragmites karka* or *Typha latifolia* or *Scirpoides holoschoenus* (as per **Figure 6**)



Typha latifolia



Scirpoides holoschoenus



Vetiver zizanioides



Canna indica



Phragmites karka

Figure 6: Plant species names



Figure 7: Sample image of a soakaway (e.g Versi Cell)

3.3.1 Advanced treatment for the food waste digester effluent

If there is any limitation or concern to discharge the treated effluent from wetland into the ground, it is recommended that a mechanical filter system is added prior to the discharge of the effluent into the wetland.

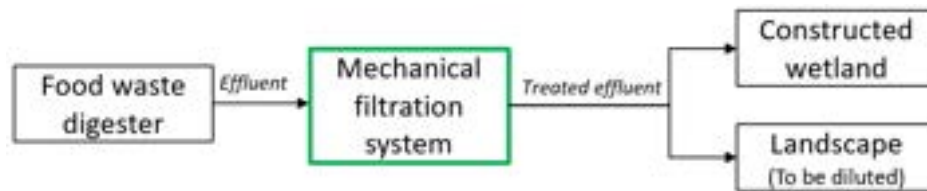


Figure 8: Process flow of the treatment for food waste digester

As per Figure 8, the process of the disposal of the food waste would first begin by being digested by the food waste digester, then being filtered by the mechanical filter system after which it can be discharged to the constructed wetland or be used to water the surrounding landscape. It is recommended that the water is diluted by a ratio of 7:1.

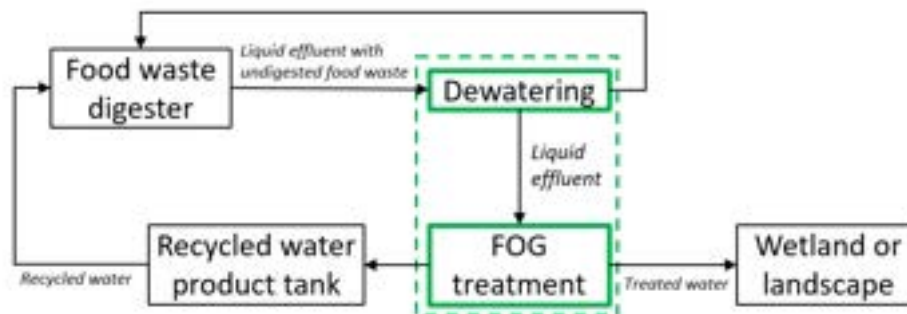


Figure 9: Process flow of the mechanical filter system

As per Figure 9, the effluent from the food waste digester is to be passed to the mechanical filtration system. It first passes through the dewatering component where solids are separated from the liquid. It then passes through the FOG treatment (fats, oil and grease) segment where fats, oil and grease are removed from the liquid. Lastly, the treated water can then be used for landscaping or be passed to the constructed wetland.

The cost of this equipment is approximate \$ 14,500 USD (i.e., \$ 20,000 SGD). To ensure optimum performance of the filter, it is recommended that the filter used is sourced from the same supplier as that of the food waste digester.

3.3.2 Alternative solution for food waste digester effluent

If it is not permitted to discharge the treated water into the ground, the effluent can be evaporated in the open field by utilising solar heating. However, a large area (approximately 750m²) is needed to be exposed to the sky to evaporate the daily generation of 6m³ of effluent from the food waste digester.

3.4 Solution proposals for excessive foaming

3.4.1 Option 1

The 1st option would be to pass the wastewater to a “pre-filter” before passing it to the equalisation tank along with the domestic wastewater (blackwater). This would ensure the foams and suds are filtered before passing it into the equalization tank as per Figure 10.

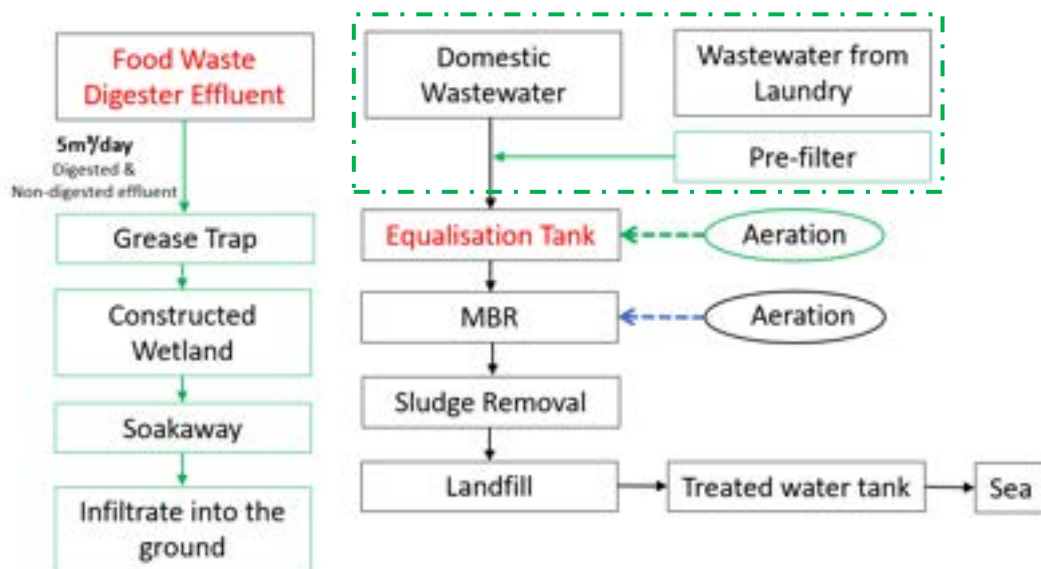


Figure 10: Treatment train to accommodate wastewater from laundry (Option 1)

3.4.2 Option 2

The 2nd option to filter the laundry wastewater and then released into the sea/ infiltrated into the ground. This is as shown in **Figure 11**.

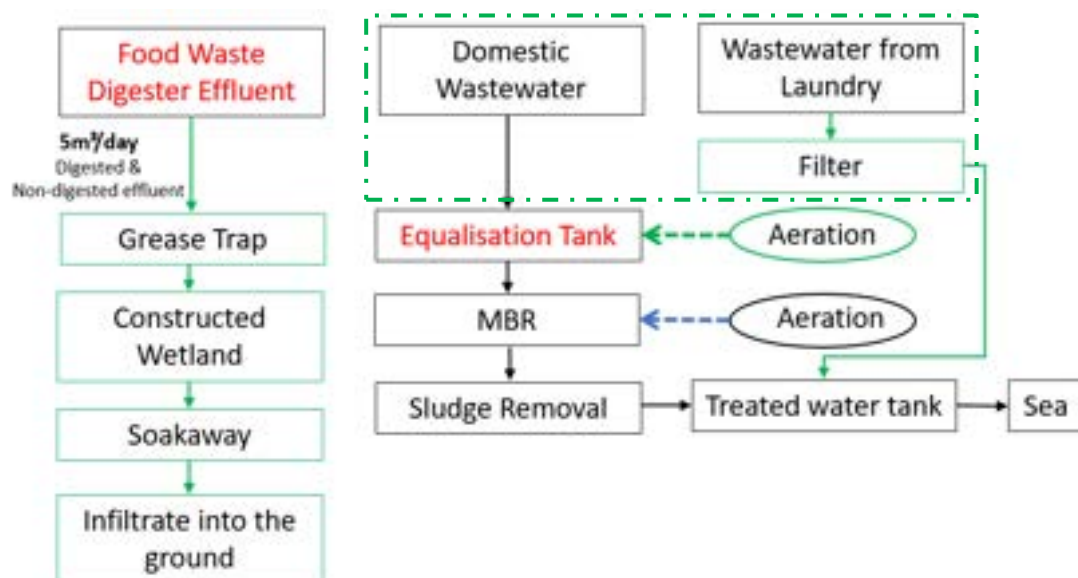


Figure 11: Treatment train to accommodate laundry water (Option 2)

3.5 Long term proposal

To ensure a sustainable wastewater management plan is in place, it is recommended that the following proposals as per **Figure 12** are given its due consideration.

- Installation of a food waste composter: Sludge from the MBR can be fully digested and processed into a solid fertiliser which can be used for landscaping and farming purposes within the resort's nearby vicinity. Food waste can be fully digested without the need of manual separation and handling of digested food waste effluent.

- Ultrafiltration treatment: Using an ultrafiltration system to further process the effluent treated water tank which can then be used for staff toilet flushing and other domestic uses such as plant watering, washing etc. This will reduce the demand for potable water.

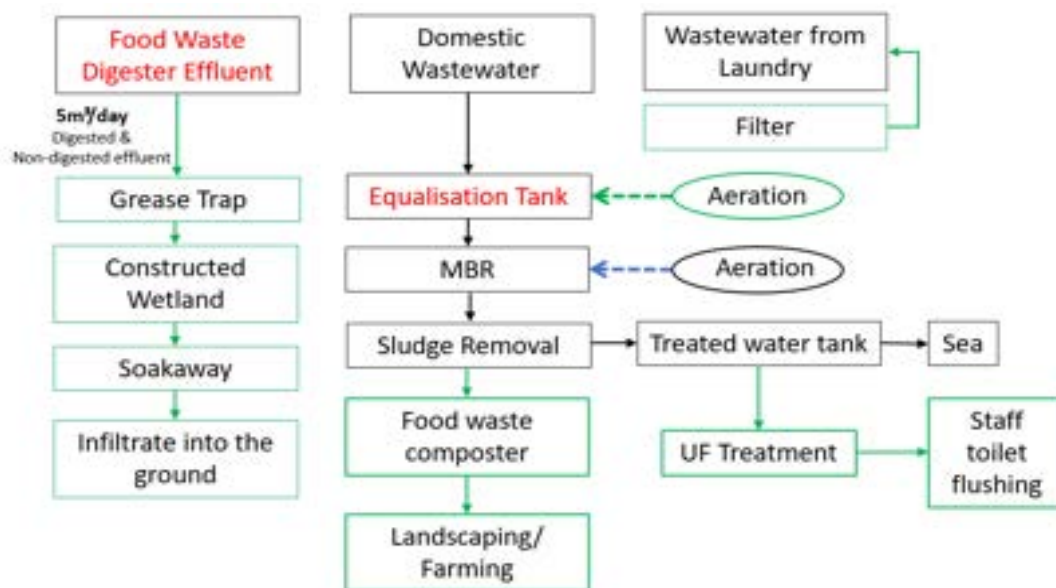


Figure 12: Overall future proposal for wastewater treatment train

4 Method statement

4.1 Intermediate solution proposal for odour issue

Risks & Control measures

To ensure a safe working environment during the execution of the intermediate solution for the odour issue faced (as detailed in **Section 3.1**), several operational health and safety hazards needs to be managed. These are as follows in below **Table 2**.

Hazards	Control measures
Gases from the equalization tank may be flammable when heat and vibration is introduced into the system during coring works	Use oxidizer to oxidize the wastewater in the oxidizing tank and ensure vent pipe is installed to ensure remaining gases are released
LIMIN8/ Potassium Permanganate may be harmful if swallowed and can cause skin and eye irritation	Wash with soap and water, seek medical attention if irritation persists

Table 2: Hazards and control measures for intermediate solution

Description of work

The follow steps detail the steps required to undertake the intermediate solution as detailed in **Section 3.1**.

1. Reduce the water level to half in the equalization tank.

2. Add oxidizer (LIMIN8/ liquid Potassium Permanganate) into the equalization tank. A pre-determined dosage is to be provided.
3. Allow for 1-2 hours for oxidization to occur and gases to release through the vent pipe.
4. Core the openings at the 4 distinct locations as illustrated in **Figure 3** in **Section 3.1**; the diameter of the openings is to be slightly larger than the size of the inlet aeration pipe – 150mm.
5. Connect the flexible PVC pipes (5m) for aeration into the equalization tank through the openings made. The pipes are to be lowered into the body of the water in the tank and to the bottom of the tank as per **Figure 13** below.
6. Connect the air blower pipe of MBR into the new pipe.
7. Monitor the condition of the smell and the air flow rate of the aerator.

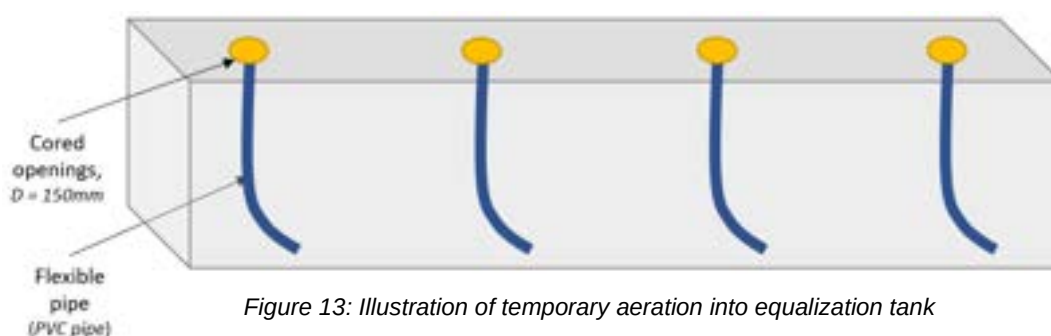


Figure 13: Illustration of temporary aeration into equalization tank

4.2 Constructed wetland for FWD

Key consideration & control measures

To ensure the constructed wetland is built based on its intended function and design, several key considerations and control measures need to be adhered to. These are as follows in **Table 3**.

Key considerations	Control measures
Defects or poor sealing between sheets of liner/ geotextile and piping	• Ensure liner/ geotextile is free from defects before installing • Necessitate checks before geotextile/ liner is overlain with filter media
Lateral bed slope varying more than 30mm from high point to low point	Ensure difference between points along bed slope is measured before filter media/ liner is placed over it
Over-compaction of substrate/ filter media/ bed soil during construction due to high foot traffic and small machine traffic on site	• Walk boards to be placed for foot traffic during planting or necessary assessments • Ensure low-ground pressure equipment is used
Animals burrowing into dikes or obstructing discharge pipes	Riprapping upstream and downstream slopes from wetland
Stormwater runoff overloading system	Diversion of stormwater runoff away from constructed wetland site
Excessive growth of weeds and other plants	Weeding to be carried out periodically to avoid the impairment of CW media and plants

Table 3: Key considerations & control measures for constructed wetland

Description of work

A. Pre-construction work

1. Preparation of plants by pre-planting or culturing shoots of selected plant species.
2. Preparation of sizing and quantity of filter media required.
3. Grease trap to be installed upstream of influent FWD effluent.

B. Construction work

1. Excavation of CW site by a depth of 0.6m, length of 7m and width of 3.5m; excavation of soakaway site by a depth of 1.0m, length of 2m and width of 3.5m.
2. Side slopes and lateral bed slope of 1:2 and 1:100 are to be carved out respectively.
3. Perforated pipes are installed into the CW site and into the soakaway.
4. CW bed soil to be lined with 12mm plastic liner; soakaway to be lined with appropriate geotextile. Care should be taken to ensure overlaps/ butted joints and anchorages are in accordance with the manufacturer's specifications.
5. Filter media is filled into the excavated site; detention tank lowered into the excavated soakaway site.
6. Shoots are planted on the CW soil (excluding the plant free zone) with the appropriate spacing of 20cm × 20cm.

C. Post-construction work

1. Gradually increasing the volume and organic load into the system by using the following watering schedule outlined in **Table 4**.
2. Water samples are to be collected and examined using an appropriate method and laboratory equipment. The water quality parameters of BOD, TSS, and colour of the effluent are to be observed/ measured and documented.
3. The plant's growth and observable physical characteristics (i.e., leaf length/ width, plant height, leaf colour, stem colour are to be observed and recorded on a periodic basis (e.g. once in 3 days). Should any signs of distress be observed on the plants during the watering schedule, the CW media and plants are to be examined with further detail.

	Day	Total Volume (m ³)	Concentration (% FWD)
Week 1	1	1	0 (Water)
	2	2	0 (Water)
	3	3	0 (Water)
	4	4	0 (Water)
	5	5	10
	6	4	10
	7	5	20
Week 2	1	4	20
	2	5	30
	3	4	30
	4	5	40
	5	4	40
	6	5	50
	7	4	50
Week 3	1	5	60
	2	4	60
	3	5	70

	4	4	70
	5	5	80
	6	4	80
	7	5	90
Week 4	1	4	90
	2	5	100
	3	4	100
	4	5	100

Table 4: Volume and organic load for constructed wetland

5 Cost structure

Based on the proposed solutions, the costs for the temporary works will be estimated based on the local conditions and given data.

6 Conclusion

Four Seasons Landaa Giraavaru faces 3 main issues which are odour, excessive foaming, solid dewatering. The odour issue can be addressed in the short-term and in the long-term by executing an intermediate solution (through increased aeration and the use of a liquid oxidizer) and a permanent solution by passing the digested food waste effluent into a constructed wetland.

There are two options to address the excessive foaming issue. The first option would be to introduce a pre-filter to the laundry water and then passing the pre-filtered laundry water into the STP – MBR. Alternatively, the laundry water can be filtered and be reused for non-potable uses such as toilet flushing.

With a view for a fully sustainable wastewater management system, it is hoped these proposals are given due consideration. The use of a food waste composter to process the sludge from the MBR and an ultrafiltration system to process the treated water is also highly recommended to further improve the existing waste management plan in Four Seasons Landaa Giraavaru.

As noticed, we recommend for a complete water balance study and sustainable water management to reduce the desalinated water demand, water treatment maintenance cost and energy consumption of the resort.

7 Appendix



WEB EARTH

1. Food Waste Digester

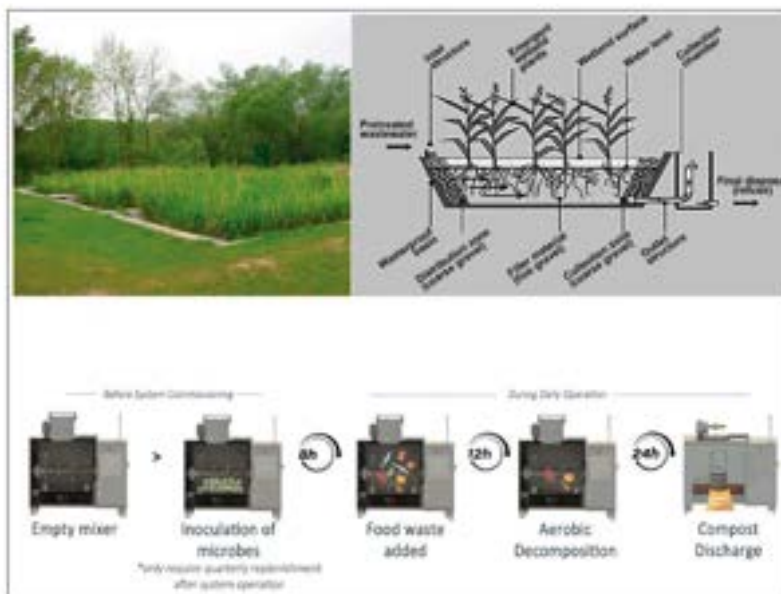
Process:

- Collection
- Segregation
- Digestion
- Liquid effluent to STP

Man power: 3 pax per day

Observation: Undigested effluent discharge at anaerobic condition

No grease trap



WEB EARTH

1. Food Waste Digester

Web Earth proposal:

1. Stop effluent discharge into STP
2. Treat effluent by constructed Wet land
3. If possible cancel the PO and buy Food waste composter – output dry compost



WEB EARTH

2. Sewage Treatment Plant

STP system: MBR

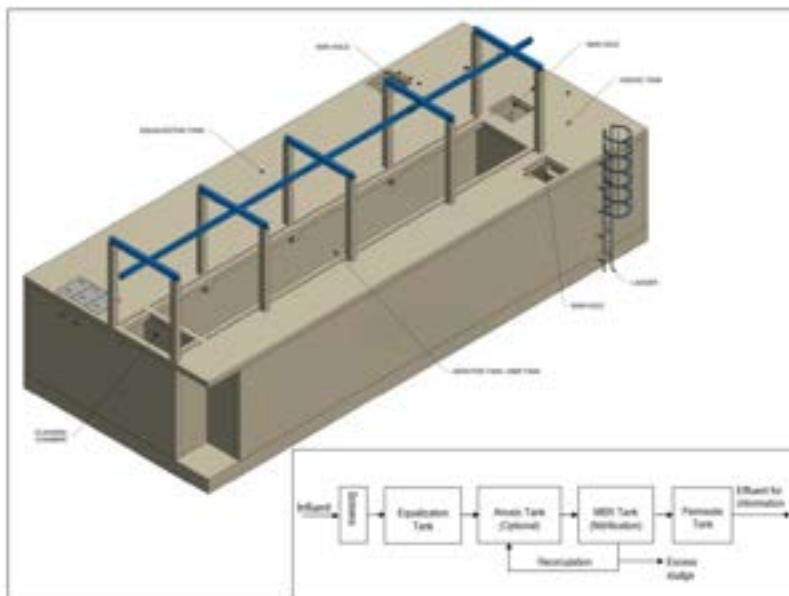
Capacity: 400m³/day

Installed: 2018

Maintenance: By FS Eng team,
limited support by supplier

Smelling issue: since day 1 but
smell so bad after food waste
digester (FWD) effluent discharge

Sludge removal:
supplier information 3bags /week
Actual before FWD 3bags/day
Current – unable to separate
activated sludge



WEB EARTH

2. Sewage Treatment Plant

Observation:

- Smell from Equalization tank
- Equalization tank at septic condition
- Poor aeration & enclosed tank
- Bad smell when water flow through screener
- No smell on MBR
- Forming issue by laundry water



WEB EARTH

2. Sewage Treatment Plant

Observation:

Forming issue by laundry water

Laundry water usage:

25-30m³/day

Current action: apply anti-form

Web Earth Proposal:

-Stop laundry water discharge into STP

-Treat laundry water by membrane filtration and reuse



WEB EARTH

4. Other Water Related Issues

Desalination process

Capacity: 500m³/day

Water demand: 400-500m³/day

Issues:

-Membrane replacement every 6months

- Scaling on water supply pipe
- Clogging kitchen waste water pipe

Annex B

Food Waste Composter

Assessment of various products



Four Season Landaa Giraavaru, Maldives

Food Waste Composter

Assessment of various products

Ref: Web(E)/108/01/FWCS/GG

Date: July 2022

1 Summary of assessed food waste composter

Company	Options from Holiday Inn - Kandooma					WEB Earth						
	Partner company of Prestige food	Watson Pvt Ltd	Integral Pvt Ltd	PowerKnot	Rendisk	ZWaste	Westcom	ECOBOT	MAECO	A-Smart	Ecogreen	Waste transformer
Product	Garden compost bins	Composter - EcoFast Italia	Integral	LF - 500	Circular	ZW-CS 500kg	WF-300kg	500kg composter	500kg composter	Smart Geo 700kg	Composter	600kg digester
Capacity (kg)	200L	N/A	400 kg	500kg	500kg	500	300	500	500	700	500kg	
Fabrication location	Australia	Maldives	Maldives	USA	Netherlands	Singapore	Singapore/ China/ Malaysia	India	Singapore	Singapore	Singapore	Netherlands
Type of treatment	Natural processing	Multi-stage system	High temperature aerobic composter	Aerobic biocomposting	Multi-stage system	Aerobic biocomposting	Aerobic biocomposting	High heat aerobic biocomposting	Heating & fermentation composting using enzymes	Aerobic fermentation with pre-treatment	Aerobic biocomposting	Anaerobic digestion
Equipment cost	USD 50	USD 18,000	USD 23,500	Not provided	Not provided	USD 64,444	USD 61,185 (300kg)	USD 46,643	USD 75,704	USD 108,000	USD 51,900	USD 314,000
Lease cost and details	N/A	N/A	N/A	N/A	N/A	> Lease to own > \$6,519 USD/ month > 1 year lease	> Option to own > \$4,471 USD/ month > 3 year lease	N/A	> Option to own > Monthly cost \$3,666 USD > 1 year with option to extend	N/A	N/A	N/A
Feeding of waste method	Batch system	Continuous	Not provided	Continuous	Continuous	Continuous	Continuous	Batch	Batch	Continuous	Batch	Continuous
Processing time	Minimum 2 months	45-60 days	24-36 hours	5-6 hours	Not provided	24 hours	24 hours	24 hours	24 hours	5 days	24 hours	Unspecified but >24 hours
Cost source	Online retail prices	Quote	Quote	N/A	N/A	Quote	Quote	Quote	Quote	Quote	Quote	Quote
Space requirements	Large space requirement; outdoor area	Large space requirement	Not provided	Not provided	> Waste station - 1 sqm > Composter - 2.5 sqm	2.7m X 1.2m	4.1m X 1.6m	3.7m X 2.6m	2.5m X 2.5m	Estimated 40-50 sqm for digester, water treatment and scrubber	2.5m X 1m	> 30 sqm
Energy consumption & power requirements	0	Not provided	720 - 1200 kWh/day (25 - 30 kW)	15 kWh/day (2.8 kW)	Processing hours not provided (15 kW)	125 kWh (5-8 kW)	60 kWh (3-5 kW)	150-240 kWh (Max. 15-17 kW)	100 kWh (35 kW)	126 kWh (0.18 kWh/kg)	288 kWh (12 kW)	30 kWh (1-2 kW)
Operating temperature	0	Not provided Should be low as no direct heating involved	65-75 CG	50 CG	Not provided	60 CG	40 CG	100-150 °C	60-85 CG	50-60 CG	60-70 CG	TBC
Liquid effluent	Yes	Yes	No	Yes	No	No (Vapour/ condensate only)	No	No (Condensate only - between 100-200L; can be used for non-potable needs)	No (Condensate only; can be used for non-potable needs)	Yes (With dry, fine compost produced)	No	Yes (Significant quantity)
Maintenance requirements	Yes, manual removal, drying and etc. Required to be buried into ground	Manual removal of compost; curing required	Not provided	Not provided	Not provided	Periodic maintenance required for cleaning and greasing	Periodic maintenance required for cleaning and greasing	On-site maintenance provided by Maldives team	Periodic maintenance required for cleaning and greasing	Required	Required	Periodic maintenance required
Key features	N/A	Requires 2-3 separate machines, space to store curing racks for composting for as long as 60 days	N/A	N/A	N/A	> Additional dual drum system & finer grade grinder at final stage > No feedback mechanism; manual adjustment required > Can add horticultural waste	> Feedback mechanism to ensure dry compost > No horticultural waste input	> Operating at high temperature ensuring dry compost > Microbes can be reused from output compost > Feedback mechanism to ensure dry compost > Excess compost can be collected by ECOBOT > Can add horticultural waste	> Optional usage of ozone filter to reduce odour > Feedback mechanism to ensure dry compost > Can add horticultural waste	Additional WWT system to treat water to allow for reuse for digester		Able to generate average of 75 kWh/day of electricity
Warranty	0	N/A	1 year	3 years	N/A	1 year	1 year	1 year	1 year	N/A	N/A	18 months
WEB Comment 1 July	Unfeasible solution - Smell and attracts pests and insects - Liquid effluent from natural biodegradation - High space requirement	Not recommended - Complicated operations - require thorough coordination - Long processing time for compost (up to 60 days for curing compost)	Not recommended - Very high energy consumption - Insufficient information	Currently in use at FSR Landaa - Require treatment for digested liquid effluent	Not recommended - Complicated operations (require thorough coordination) - Long processing time for compost (up to 60 days for curing compost)	Satisfactory - Manual feedback mechanism	Can be considered if customisation made: - Cannot handle horticultural waste - Large quantity of microbes required	Satisfactory - Low capital cost - Local maintenance and support available - Short lead time	Satisfactory - High capital cost	Can be considered if customized and can have large area. - Liquid effluent need to be treated to allow for reuse for digester - Requires large space for digester, water treatment, gas scrubber	Not recommended - No track record - No grinder	Not recommended - High capital cost - No track record - Operational risk if system leaks

Table 1: Summary of 12 food waste composters from various vendors

2 Summary of shortlisted food waste composter

Company	WEB Earth			
	ZWaste	Westcom	ECOBOT	MAECO
Product	ZW-CS 500kg	WF-300kg	500kg composter	500kg composter
Capacity (kg)	500	300	500	500
Fabrication location	Singapore	Singapore/ China/ Malaysia	India	Singapore
Average lead time (months)	3	3.5	1.5	2.5
Type of treatment	Aerobic biocomposting	Aerobic biocomposting	High heat aerobic biocomposting	Heating & fermentation composting using enzymes
Bin lifter, load cell & grinder?	Yes	Yes	Yes	Yes
Equipment cost*	USD 64,444	USD 61,185 (300kg)	USD 46,643	USD 75,704
Lease cost and details	> Lease to own > \$6,519 USD/ month > 1 year lease	> Option to own > \$4,471 USD/ month > 3 year lease	N/A	> Option to own > Monthly cost \$3,666 USD > 1 year with option to extend
Microbes/ enzyme requirement per load	N/A	N/A	1:5000	1:1000
Yearly cost of microbes	USD 2,296	USD 9,600	USD 185	USD 2,982
(Cost of composter + 5-year microbes) / Total waste	USD 0.08 /kg	USD 0.21 /kg	USD 0.05 /kg	USD 0.1 /kg
Feeding of waste method	Continuous	Continuous	Batch	Batch
Processing time	24 hours	24 hours	24 hours	24 hours
Space requirements	2.7m X 1.2m	4.1m X 1.6m	3.7m X 2.6m	2.5m X 2.5m
Energy consumption & power requirements	125 kWh (5-8 kW)	60 kWh (3-5 kW)	150-240 kWh (Max. 15-17 kW)	100 kWh (35 kW)
Operating temperature	60 °C	40 °C	80-150 °C	60-85 °C
Liquid effluent	No (Vapour/ condensate only)	No	No (Condensate only - between 100-200L; can be used for non-potable needs)	No (Condensate only; can be used for non-potable needs)
Average compost output capacity	175 kg	30 kg	150 kg	200 kg
Maintenance requirements	Periodic maintenance required for cleaning and greasing	Periodic maintenance required for cleaning and greasing	On-site maintenance provided by Maldives team	Periodic maintenance required for cleaning and greasing
Key features	> Additional dual drum system & finer grade grinder at final stage > No feedback mechanism; manual adjustment required > Can add horticultural waste	> Feedback mechanism to ensure dry compost > No horticultural waste input	> Operating at high temperature ensuring dry compost > Microbes can be reused from output compost > Feedback mechanism to ensure dry compost > Excess compost can be collected by ECOBOT > Can add horticultural waste	> Optional usage of ozone filter to reduce odour > Feedback mechanism to ensure dry compost > Can add horticultural waste
Warranty	1 year	1 year	1 year	1 year
Refer Section 3 - Conclusion for more details)	Satisfactory - Manual feedback mechanism	Can be considered if customisation made: - Cannot handle horticultural waste - Large quantity of microbes required	Satisfactory - Low capital cost - Local maintenance and support available - Short lead time	Satisfactory - High capital cost

Table 2: Shortlisted food waste composter (ZWaste, Westcom, ECOBOT, MAECO)

3 Conclusion

Among the 12 assessed composters, 4 composters were shortlisted after various engagements with the respective vendors.

The shortlisted composter vendors include ZWaste, Westcom, ECOBOT and MAECO. The composters were shortlisted as the product offered met the basic functional requirements for FSR Landaa Giraavaru site. The following key features were considered during the assessment of the shortlisted composters.

- Dry compost out product & no liquid effluent
- Capital, consumables and operational cost
- Maintenance support
- Accessibility & easy operation
- Space requirement

The advantages of **ZWaste's** 500kg composter includes its dual-drum system which allows continuous feeding of waste, an in-built grinder and it is suitable for horticultural waste input. Limitations of this composter is the absence of automatic adjustment of the operational parameters. ZWaste composter's total 5-year cost (capital cost + 5 years microbes) per total waste fed is \$0.09USD/kg.

The advantages of **Westcom's** 300kg composter are that it has its own R&D facility for bacteria cultivation, allows for continuous feeding and automatically adjusts operational parameters to produce dry compost. To fulfil the short fabrication time required, only a 300kg composter with an in-built grinder is recommended. Other limitations include limited quantity of horticultural waste can be added (only leaves) and high yearly tonnage microbes required (10 ton/year) – incurring high transport costs. Westcom's 300kg composter's total 5-year cost (capital cost + 5 years microbes) per total waste fed is \$0.21USD/kg.

The advantages of **MAECO's** 500kg composter are that it has its own R&D facility for its enzyme cultivation, suitable for horticultural waste input and automatically adjusts operational parameters to produce dry compost. The limitations of the composter are that it does not allow continuous feeding of waste. The capital & yearly maintenance cost are high compared to other 3 shortlisted vendors. The total 5-year cost (capital cost + 5 years microbes) of the composter per total waste fed is \$0.1USD/kg.

The advantages of **ECOBOT's** 500kg composter are that it has a local support team available with access to spare parts, low capital cost, in-built grinder, suitable for horticultural waste input, automatically adjusts operational parameters to produce dry compost, has high operating temperature with insulating jackets to maintain safe temperature of the drum while ensuring dry compost is produced, provides service to distribute excess compost to regional farmlands in Maldives, low microbes cost – no requirement to purchase microbes as dry compost can be reused as a medium for microbes. The limitation of the composter is that it does not allow continuous feeding of waste. The total 5-year cost (capital cost + 5 years microbes) of the composter per total waste fed is \$0.05USD/kg.

Based on the above assessment we recommend ECOBOT and MAECO as both fulfils almost all the requirements. Though, ECOBOT is preferred as it provides local technical support, offers compost collection and has lower capital and operational cost. The final decision should be based on the feedback from site visits to both ECOBOT and MAECO composters respectively.

4 Appendix

1. ZWaste



Figure 1: ZWaste Composter

A) Financing for Composter	500kg composter	1000kg composter
Total Cost	Outright purchase – S\$87,000 Est. 120L Bins – S\$2,000	Outright purchase – S\$125,000 Est. 120L Bins – S\$2,000
Leasing options (if available) Upfront cost & leasing period	Lease-to-own – 1 year – S\$8,800 per month 3 years – S\$3,600 per month 3 months security deposit	Lease-to-own – 1 year – S\$12,600 per month 3 years – S\$5,200 per month 3 months security deposit
Obligations at the end of contractual period (if any)	One month leasing fee to own the system	
Cost of microbes per year (based on needs of composter)	Outright purchase – S\$3,100 per year Lease-to-own - Inclusive	Outright purchase – S\$5,200 per year Lease-to-own - Inclusive

Figure 2: Details of ZWaste composter (1)

B) Conditions & warranties	500kg composter	1000kg composter
Period of warranty & coverage	Outright purchase – 1 year Lease-to-own – Throughout leasing period Coverage – Manufacturing defects (Parts and Labour)	
Other relevant conditions or key innominate terms of agreement	Refers to separate leasing agreement template	

Figure 3: Details of ZWaste composter (2)

C) After sales services	500kg composter	1000kg composter
Provision for in-person technical support (ad-hoc and periodic basis)	Option 1 – ZWS Maldives technical representative Option 2 – Hotel engineering team to be trained by ZWS	
Provision for virtual/ remote technical support (ad-hoc and periodic basis)	Inclusive – As and when needed	
Training of project team (based in Maldives)	Inclusive for 1 st installation	
Installation supervision and commissioning	Inclusive for 1 st installation	

Figure 4: Details of ZWaste composter (3)

E) Lead time and logistics	500kg composter	1000kg composter
Estimated lead time	12 – 14 weeks	
Mitigation plans in the event of supply chain/ logistical constraints (if any)	Critical components – Motor – Stock SS Fabrication – Singapore HMI & PLC - Stock	

Figure 5: Details of ZWaste composter (4)



Figure 6: Client reference list

REFERENCE LIST - MALDIVES					
Owner	Sites	Type	Capacity	Year	Remark
Singha Group - Thailand	Hard Rock Hotel	Composter	1ton	2019	
	Curio Hotel	Composter	1ton	2019	
	Main BOH	Composter	1ton	2019	
Pontiac Land Group - Singapore	Rah Falhu Hura Resort - Island 1 Park Hyatt	Digester + Composter	Digester – 150kg Composter – 500kg	2019 – 2020	Digester used during construction stage for site food waste
	Rah Falhu Hura Resort - Island 3 Ritz Carlton	Composter	300kg	2020	
	Rah Falhu Hura Resort - Island 4 Staff Campus	Composter	500kg	2020	
Chiu Teng Group - Singapore	Le Meridien Resort	Composter	300kg	2019	

Figure 7: ZWaste client reference in Maldives

2. Westcom



Figure 8: Westcom composter

Item	Qty	Unit	Product Description	Unit Price SGD	Total SGD
1.1	1	unit	Westcom WF-300 BioDigester - Onsite Aerobic Treatment Process - Dry discharge System, no water piping - 24 hours food waste degradation to fertilizer - Continuous process - Smart Start Stop sensor, calibrated to low working temperature - MicroBites® Bio safety level 1 - Treatment capacity 300kg	\$69,000.00	\$69,000.00
1.2	1	yearly	1-year Monthly Service and Maintenance - Microbes top up - Consumables (RFID up to 10 Users) - Monthly machine servicing - Harvest collection *Excludes spare parts	\$12,960.00	\$12,960.00
					\$81,960.00

Prices:	All prices in SGD Dollars, excluding GST, valid for 30 days
Purchase Payment terms:	30% Deposit upon order confirmation, balance 70% upon delivery
Delivery:	5-6 months after receipt of deposit
1 time Deployment per site *	Local Transportation, commissioning, 1 x Training and installation - \$1800
Leasing Contract	Deposit Upfront 3+1 months
Purchase Option after lease	Option to purchase unit(s) based on lease price of 1 month/unit with zero warranty coverage

*All cost listed are estimated, other additional costs like integrated systems and architectural installations, are not inclusive in this quotation. Additional power supply, supply and install cables and cables support system, \$150 per meter.

Figure 9: Cost quote for Westcom

A	Financing for composter			
1	Total cost	\$82,600	\$102,600	\$150,000
2	Leasing options (if available) Upfront cost & leasing period with Spares and local labour.	\$6,036 per month/ 3 years, Minimum 1 year leasing period	\$7217 per month/ 3 years, Minimum 1 year leasing period	\$9092 per month/3 years, Minimum 1 year leasing period
3	Obligations at the end of contractual period (if any)	Option to purchase after leasing period ends	Option to purchase after leasing period ends	Option to purchase after leasing period ends
4	Cost of microbes per year (based on needs of composter) reference to A1	\$12,960	\$16,800	\$21,600
B	Conditions & warranties			
1	Period of warranty & coverage - reference to A1	1 year, part exclude labour	1 year, part exclude labour	1 year, part exclude labour
2	Other relevant conditions or key innominate terms of agreement	TBA	TBA	TBA
C	After sales services			
1	Provision for in-person technical support (ad-hoc and periodic basis)	\$400/4hours (local)	\$400/4hours (local)	\$400/4hours (local)
2	Provision for virtual/ remote technical support (ad-hoc and periodic basis)	\$400/4hours for Trial Unit	\$400/4hours for Trial Unit	\$400/4hours for Trial Unit
3	Training of project team (based in Maldives)	TBA	TBA	TBA
4	Installation supervision and commissioning	\$6000/deployem nt (local)	\$6000/deployem ent (local)	\$6000/deployment (local)
E	Fabrication			
1	Location of fabrication site (i.e., factory)	Singapore/China/ Malaysia	Singapore/China / Malaysia	Singapore/China/ Malaysia
2	Location of microbes cultivation facility (i.e., laboratory etc.)	Delivery from Singapore	Delivery from Singapore	Delivery from Singapore
F	Lead time and logistics			
1	Estimated lead time	1 unit Ex Work Singapore in November 2022	6 months for first 2 units, balance TBA upon final confirmation	6 months for first 2 units, balance TBA upon final confirmation
2	Mitigation plans in the event of supply chain/ logistical constraints (if any)	TBA	TBA	TBA
G	Track record (> 300 kg)			
1	Years in operation & track record	3 years	3 years	3 years

Figure 10: Comparison of different composter capacity from Westcom

3. ECOBOT



Figure 11: Ecobot composter



Figure 12: Allowable items for processing

A Financing Options for Composter

1 Total CAPEX Cost* – CIF Male' Port 250KG USD 27,596, 500KG USD 41,540 1000KG USD 62,060

2 Leasing Options (if available) - not at this time

3. Obligations at the end of the contractual period (if any) - n/a

B Consumables required (i.e. microbes)

1 Cost of microbes - 01 years stock 500KG 37KGS FOC 1000KG 74 KGS FOC

if required then USD 5 / KG

C Conditions and warranties

1 Period of warranty coverage - 01 Warranty against manufacturing defects. Replacement parts will be provided in case of any failure during warranty on receipt of damaged parts.

2 Extent of coverage - The following damage will not be covered under warranty clause:

Damage to shredder or any other part due to feeding of hard material including metals, stones, large branches/ wooden articles, hard plastics, glass or ceramic ware or any other inorganic material

Damage to any electrical part due to voltage fluctuation

Damage to any electrical part due to water

Any damage due to misuse or manhandling of equipment

3 Other relevant conditions or key innominate terms of agreement – n/a

D After sales service

1 Provision for in-person technical support (ad-hoc and periodic basis) – 24/7 local support (any time required our local engineer will travel to site for any problems return transfers food and accommodation at site provided,

2 Provision for virtual / remote technical support (ad-hoc and periodic basis) - online via WhatsApp chat or video calls. WhatsApp group with foreign and local engineer including myself together with the resort engineering team so that we can have updates if any problems or questions arise

3. Training of project team (based in Maldives) - local or foreign engineers will be at site while installation and commissioning to give all the training required to the team

4. Installation supervision and commissioning – local or foreign engineers will travel to site for installation and commissioning

Figure 13: Summary of details for ECOBOT composter (1)

E Fabrication

- 1 Location of fabrication site (i.e factory) – Mumbai India
2. Location of microbes cultivation facility (i.e. laboratory etc) – Pune India

F Lead time and logistics

- 1 Estimated lead time required upon signing of S&P / lease agreement – 45 days for the production and dispatch
- 2 Mitigation plans in the event of supply chain / logistical constraints (if any) – n/a

G Track Record

1 Years in Operation & track record –

EcoBot – Established 2017

More than 400 machines in 18 countries

Russia, Cost Rica, Saudi Arabia (Aramco) , Indonesia, Australia, India, Maldives, Qatar, USA, UK, Italy, France, UAE, Philippines, Thailand, Nigeria, Peru, Singapore, just to name a few

Arivara Inc – Established February 2021

First Machine was installed in Kandima 750 in July 2021

Total Productions / Disptach / Installations 15 machines including 1 of Maeko

Kandima 750 & 1000, Baros 750, Milaidhoo 750, Velassaru 750, Contance Moofushi 500, Constance Halaveli 500, Angsana Velavaru 500, Nova (earlier Vakarufalhi) 500, Kuramathi 750, Huvafeufushi 750, Hih 500 250, Nautilus 250

Maeko – Joali Maldives 500

*Load Measurement Arrangement

250KG USD 1,346 | 500KG USD 1,518 | 1000KG USD 1,955

*Bin Lifter with Conveyor (drawing attached in email)

Total Setup USD 6,785

Figure 14: Summary of details for ECOBOT composter (2)

4. MAECO



Figure 15: MAECO composter

ecoLISER (Organic Fertilizer)	
Output	Organic Fertilizer
Processing Duration	Batch process (9-16hrs)
Input Format	One-Time per Batch
Heating Element	Yes
Odour Mitigation	Ozone Filter
Daily Processing Capacity	50kgs – 10T per batch
HFA Application Ratio	1kg HFA to 1ton of waste (Top up every batch)
Plumbing Requirements	* Water supply into machine: No * Water for cleaning & drainage: Yes
Wastewater Treatment Requirement	No
Support for Maintenance and repairs	Available

Figure 16: MAECO composter information

Specifications	ecoLISER-1Ton
Dimensions of ecoLISER (L x W x Hmm)	3500x1700x2200
Power rating of ecoLISER(KW)	55
Main motor power (kW)	19
Electric heater power (kW)	24
Material of digestion chamber	304 Stainless steel
Net weight of ecoLISER without raw material	2500kg
Raw material maximum input capacity per batch	1000kg
Fertiliser output capacity per ton of input	300-550kg
HFA required per batch (kg)	1kg per batch
User interface control system	Yes
Operating Temperature (°C)	60-85

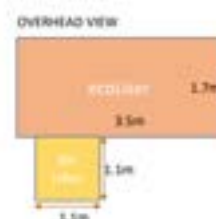


Figure 17: Technical requirements of MAECO composter

No	Essential Component	Qty	Unit Price	Amount (SGD)
1	ecoLISER 500kg System Dimension: 2500L x1400W x 1800H mm Total power rating: 35 kW Main motor power: 7.5 kW Electric heater power: 24 kW 3-phase electricity requirement: 60 Ampere Material of digester chamber: 304 Stainless Steel Net weight of ecoLISER without raw material: 750kg Raw material maximum input capacity per batch: 500kg Voltage assumption: 415V, 50Hz Inclusive of: - First year parts warranty (excluding labour). - Supervision for on-site installation & commissioning - First year training support - to be conducted in Singapore for the Client project team / service provider appointed by the Client to provide local after-sales support. 1-year supply of HFA (183kg) Leasing option: SGD4,950.00 per month Warranty against workmanship and parts defects for one year upon commissioning or 15 months upon delivery, whichever comes first. Improper use and/or installation are not warrantable. Segregation of food waste is the responsibility of Client. Please refer to terms & conditions for details.	1 UNIT	75,375.00	75,375.00
2	Bin Lifter	1 UNIT	7,980.00	7,980.00
3	Industrial weighing scale (to be integrated with bin lifter) Capacity: 150kg Readability: 0.05kg Material: Stainless Steel 304, Glass blasted finishing	1 UNIT	4,860.00	4,860.00
4	Grinder Dimensions: 1050L x 730W x 910H mm Total power rating: 0.55 kW 3-phase electricity requirement: 1 Ampere Material of digester chamber: 304 Stainless Steel Net weight of Grinder without raw material: 150kg Raw material maximum input capacity per batch: 50kg/hour Voltage assumption: 415V, 50Hz	1 UNIT	13,980.00	13,985.00
No	Other Components / Services	Qty	Unit Price	Amount (SGD)
1	1-year supply of High-efficiency Fermentation Agent (HFA)	183 KG	22.00	4,026.00

Figure 18: Details of MAECO composter costing (1)

ESTIMATED ENERGY CONSUMPTION

Following estimation is to show the difference in energy consumption based on different operating scenario (example: 16 hour operation vs 24 hour operation).

Scenario	Assumed operating conditions	Estimated daily energy consumption
1	Waste composition: 50% food waste + 50% horticulture waste Moisture reduction from 50% to 30% Total processing time 16 hours	80 kWh
2	Waste composition: 50% food waste + 50% horticulture waste Moisture reduction from 50% to 20% Total processing time 24 hours	100kWh

Figure 19: Details of MAECO composter energy consumption (2)

PROPOSED PROJECT IMPLEMENTATION SCHEDULE

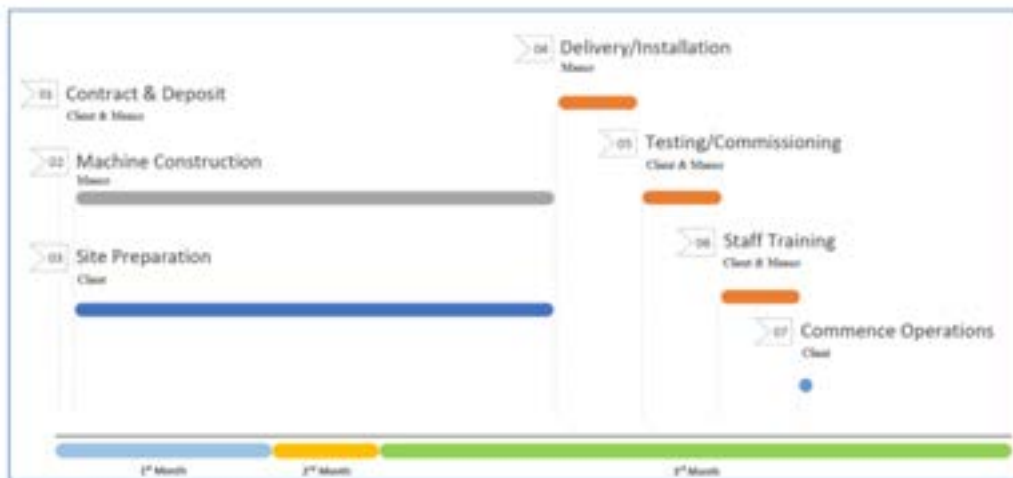


Figure 20: Details of MAECO composter timeline (3)

PROJECT TRACK RECORD

Project Name	Location	Estimated Project Value	Date Completed	Role
Food waste to fertilizer (2 ton daily)	1 Utama shopping centre, Malaysia	USD 250,000	2019	Manufacturer
Food waste to liquid (1 ton daily)	Courtyard by Marriott, Singapore	USD 100,000	2018	Manufacturer
Food waste to liquid (50kg x 10)	Changi Airport T4, Singapore	USD 160,000	2018	Manufacturer
Food waste to liquid (1 ton daily)	Crowne Plaza Hotel, Singapore	USD 100,000	2018	Manufacturer
Food waste to fertilizer (50kg daily)	Nanyang Girls High School, Singapore	USD 30,000	2018	Manufacturer
Food waste to fertilizer (50kg daily)	Yishun Primary School, Singapore	USD 30,000	2018	Manufacturer
Food waste to fertilizer (2 ton daily)	Thai Nippon Foods, Thailand	USD 200,000	2019	Manufacturer
Farm waste to fertilizer (20 ton daily)	Ao Deng Farm, China	USD 1,100,000	2018	Manufacturer & Contractor
Farm waste to fertilizer (25 ton daily)	DBN Farm, China	USD 1,300,000	2017	Manufacturer & Contractor
Farm waste to fertilizer (50 ton daily)	An Hui Chicken Farm, China	USD 2,100,000	2019	Manufacturer & Contractor
Farm waste to fertilizer (3 ton daily)	Chongqing Chicken Farm, China	USD 270,000	2019	Manufacturer & Contractor

Figure 21: Details of MAECO composter project reference (4)

5. Garden compost bins – Partner of Prestige foods



Figure 22: Garden compost bins

6. Composter – EcoFast Italia



Figure 23: EcoFast dewatering system and food waste composter

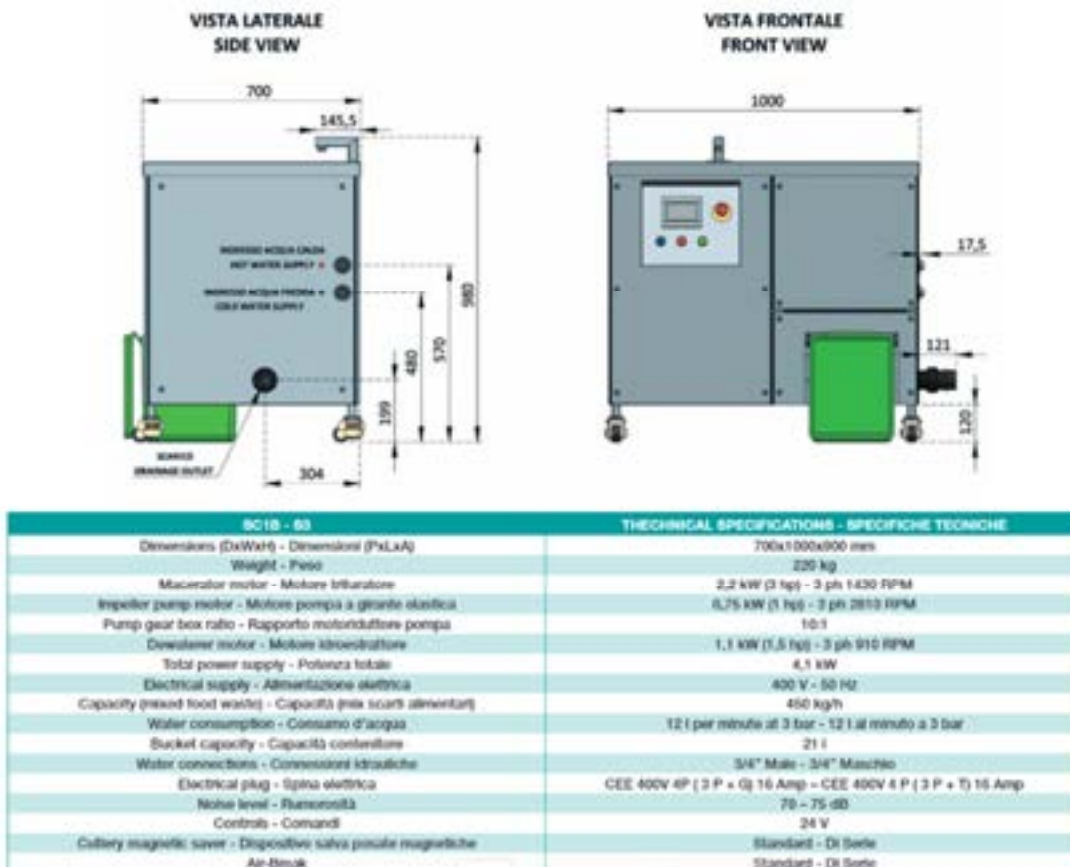


Figure 24: Power requirements for EcoFast

Reference: Food Waste Machines by EcoFast Italia Srl

Pictures	Code	Features	D mm	W mm	H mm	C	Unit Price - Ex Works - Eur
	Set up A						
	S 3	Grinding & Dewatering	700	1000	900	€	18.104,00
							-
							-
	Set up B						
	S 390 +	Grinding	546	416	900	€	6.158,00
	D 800	Dewatering	720	600	817	€	11.361,00
							-
	Set up C						
	S 651 +	Grinding	650	700	900	€	11.654,00
	D BETA	Dewatering	720	600	717	€	6.597,00
							-

Figure 25: EcoFast price list

7. Integral



Figure 26: Integral Composter

Process Flow Diagram:

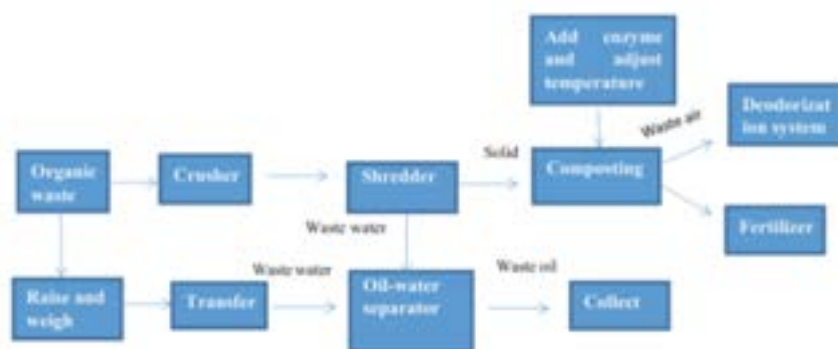


Figure 27: Process flow diagram

Technical Specification:

Processing period (h)		24/36h
Handling method:		Microbial Fermentation and Hot Air Drying
Decrement rate (%)		80/90%
Service life (h)		100000
Noise dB		65
Fermentation temperature (°C)		65-75
Power (Kw)	Solid-liquid separator (Kw)	3
25	Conveyor (Kw)	2.25
	Stirring motor (Kw)	3
	Crusher motor (Kw)	3
	Hot air blower (Kw)	0.45
	Electric heating (kw)	12
	Other	5.4
Equipment Power	Power Supply	3 phases and 4 lines *410V, 50/60Hz
	Max Power (h)	5kw
	Average Power(d)	30-50kw
of main	Appearance	Q235 + Appearance spray paint

Figure 28: Technical specification for Integral composter

9. Rendisk

	Solus Eco				Flex Wastefluge			RC-gate
	Food	Domestic	Circular		FWS	DMF	Circular	
								
Description	Our most compact solution, a stand alone unit with an extremely small footprint (0.4 m²).				The most advanced system, multiple organic waste streams, the central food waste storage, 5mm piping, various system, choice between three variants. Storage tank (FWS), bioreactor and DMF or Circular.			A refined stand alone solution suitable for tough conditions.
Dimensions (W x D x H)	600x 750x 950 mm	Footprint waste station 1 m², tank (optional) 0.4 m²	Footprint waste station 1 m², bioreactor 1 m²	Footprint waste station 1 m², Circular 0.5 m²	Footprint tank and pump 3 m²	Footprint 2 m²	Footprint 3 m²	600x 750x 950 mm
Capacity (kg per day)	1.5kg	200 – 300	200 – 300	1.5kg	1.5kg – 200kg	1.5kg – 200kg	1.5kg – 200kg	1.5kg – 200kg
Max. distance waste station and storage	NA (stand alone)	40 metres	NA (stand alone)	40 metres	1.200 metres With extra pump and additional	Circular NA, 1500 per metres		
Max number of kitchen units	NA (stand alone)	2	3	3	2	2	2	Circular NA, 1500, 1
Volume reduction	90%	NA (best waste storage only)	90%	90%	NA (best waste storage only)	90%	90%	90%
Max. Power draw	1.5kW	0.5kW	1.5kW (p-r)	1.5kW (p-r)	1.5kW	0.5kW	1.5kW	1.5kW

Figure 32: Rendisk technical specification

Annex C

A. Payback period for GLW oil filter

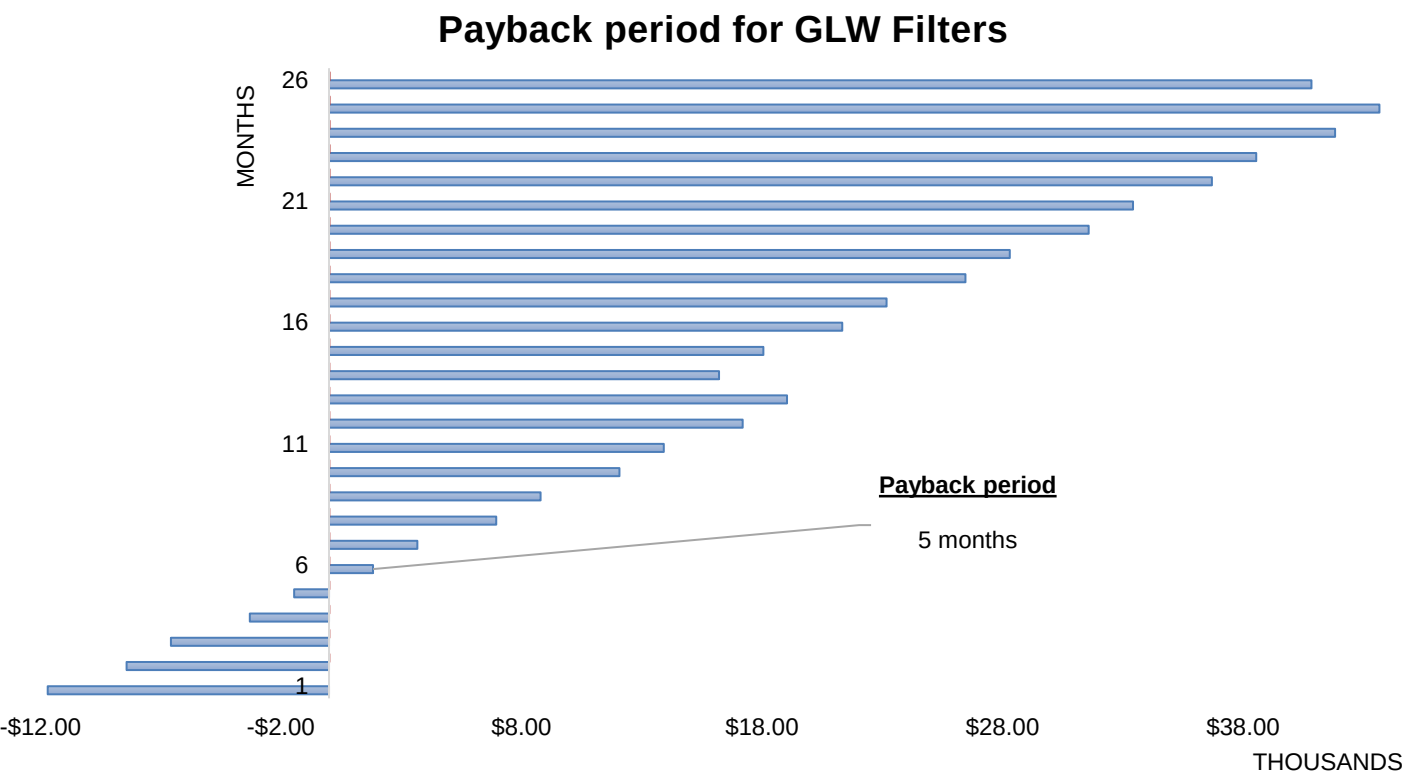


Figure B 1: Payback period for GLW filters

B. Payback period for 500kg food waste composter (ECOBOT)

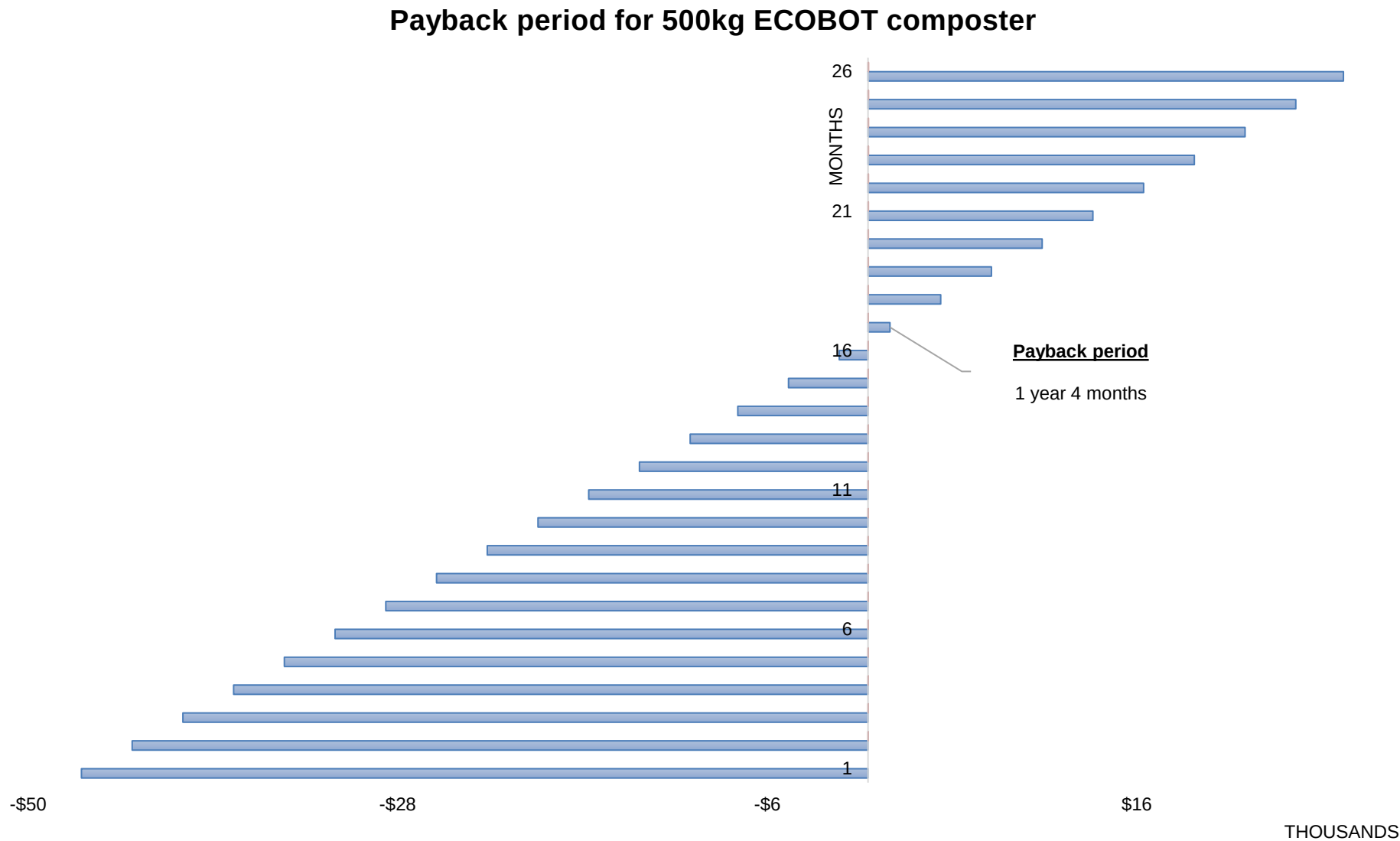


Figure B 2: Payback period for 500 kg ECOBOT composter

C. Payback period for 500kg food waste composter (MAECO)

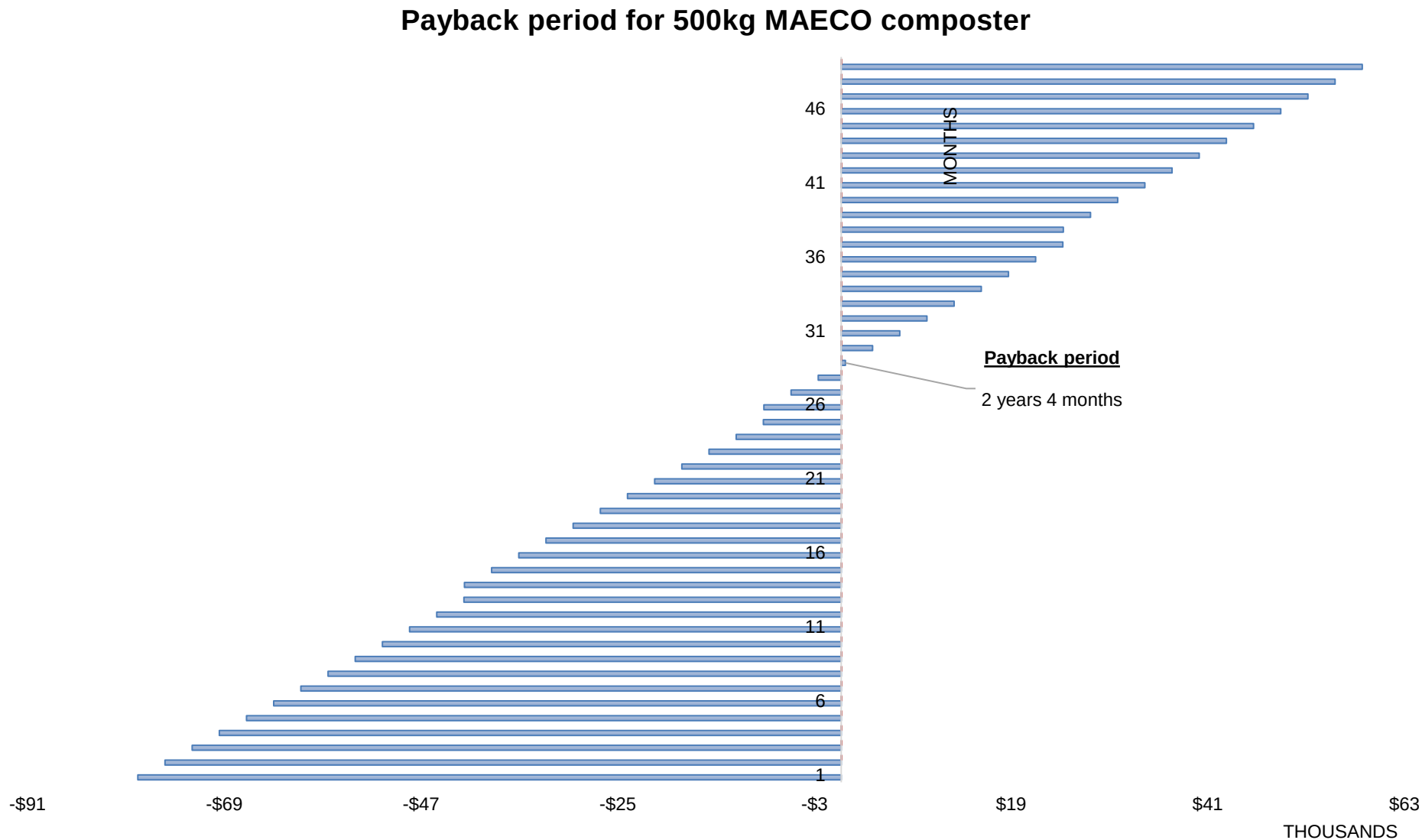


Figure B 3: Payback period for 500 kg MAECO Composter

D. Payback period for laundry water treatment system (ECOSOFT)

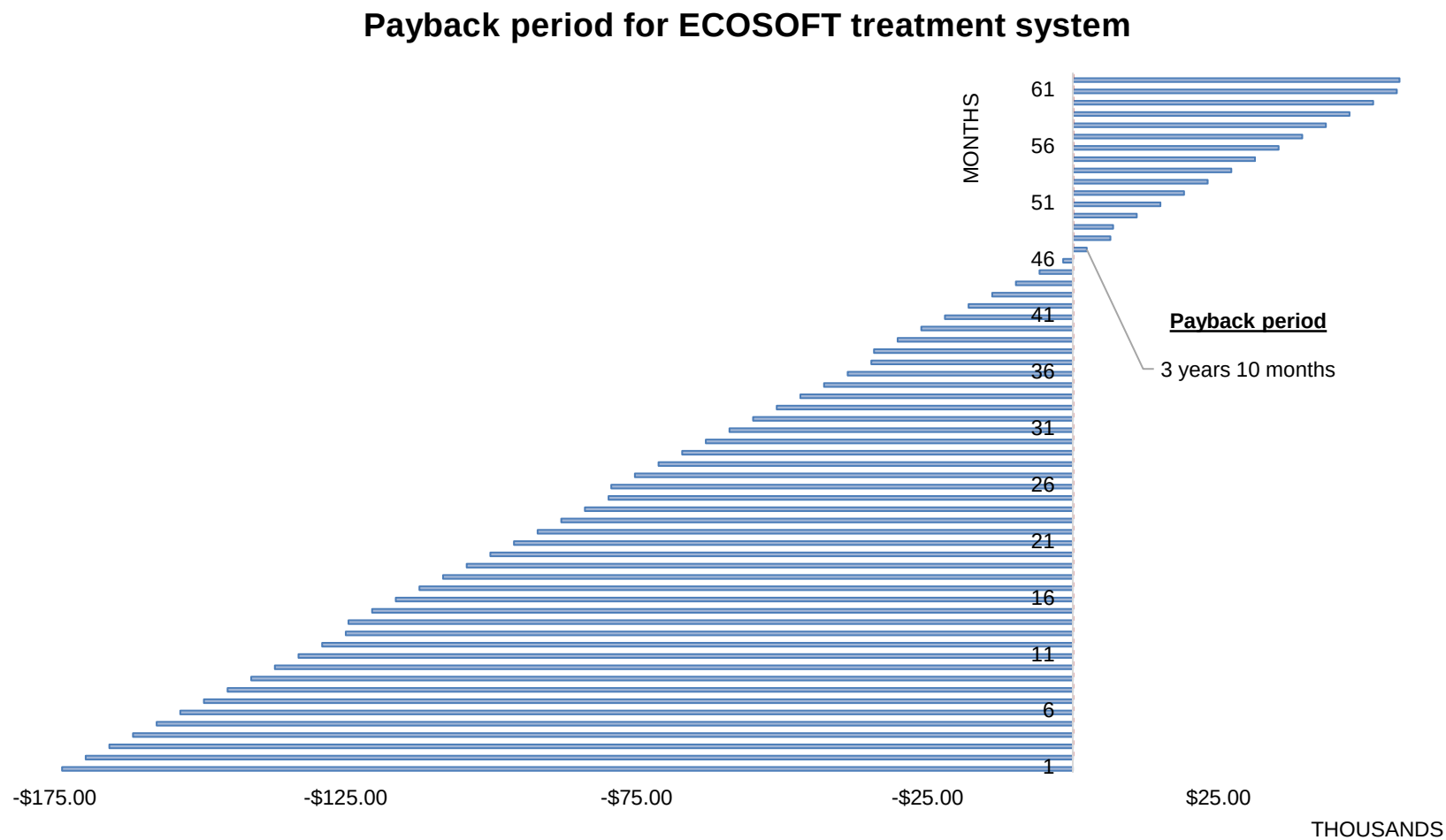


Figure B 4: Payback period for ECOSOFT treatment system

Annex D

A. Four Seasons waste tracking document

December 2021

Waste Sent to Land Fills - Track Sheet - FS LG						
Month: Dec 2021						
Week	Waste Material	Boat Trip 1		Boat Trip 2		Name
Week 1	Batteries		Numbers		Numbers	
	E-Waste		Kilogram		Kilogram	
	Oil (Kitchen)		Liters		Liters	
	Oil (Engine)		Liters		Liters	
	Food Waste		Kg		Kg	
	Mixed Waste	16.5	Tonnes	4	Tonnes	Nabir
Week 2	Batteries		Numbers		Numbers	
	E-Waste		Kilogram		Kilogram	
	Oil (Kitchen)		Liters		Liters	
	Oil (Engine)		Liters		Liters	
	Food Waste		Kg		Kg	
	Mixed Waste	4	Tonnes	16.5	Tonnes	Nabir
Week 3	Batteries		Numbers		Numbers	
	E-Waste		Kilogram		Kilogram	
	Oil (Kitchen)		Liters		Liters	
	Oil (Engine)		Liters		Liters	
	Food Waste		Kg		Kg	
	Mixed Waste	16.5	Tonnes	16.5	Tonnes	Nabir
Week 4	Batteries		Numbers		Numbers	
	E-Waste		Kilogram		Kilogram	
	Oil (Kitchen)		Liters		Liters	
	Oil (Engine)		Liters		Liters	
	Food Waste		Kg		Kg	
	Mixed Waste	4	Tonnes	4	Tonnes	Nabir
Week 5	Batteries		Numbers		Numbers	
	E-Waste		Kilogram		Kilogram	
	Oil (Kitchen)	540	Liters		Liters	
	Oil (Engine)	1000	Liters		Liters	
	Food Waste		Kg		Kg	
	Mixed Waste	4	Tonnes	16.5	Tonnes	Nabir

Total for the Month:	Total number of Batteries:	
	Total kg of E-Waste:	
	Total Liters of Oil Kitchen:	540
	Total Liters of Oil engine:	1000
	Total Tonne of Mixed Waste:	100 ton

Checked by:	Faisal
Name:	
Date:	28.12.2021
Signature:	

Figure C 1: Waste generated for December 2021

November 2021

Waste Sent to Land Fills - Track Sheet - FS LG						
Month: Nov 2021						
Week	Waste Material	Boat Trip 1		Boat Trip 2		Name
Week 1	Batteries		Numbers		Numbers	
	E-Waste		Kilogram		Kilogram	
	Oil (Kitchen)		Liters		Liters	
	Oil (Engine)		Liters		Liters	
	Food Waste		Kg		Kg	
	Mixed Waste	4	Tonnes	4	Tonnes	HARU HARU
Week 2	Batteries		Numbers		Numbers	
	E-Waste		Kilogram		Kilogram	
	Oil (Kitchen)		Liters		Liters	
	Oil (Engine)		Liters		Liters	
	Food Waste		Kg		Kg	
	Mixed Waste	16	Tonnes		Tonnes	HARU HARU
Week 3	Batteries		Numbers		Numbers	
	E-Waste		Kilogram		Kilogram	
	Oil (Kitchen)		Liters		Liters	
	Oil (Engine)		Liters		Liters	
	Food Waste		Kg		Kg	
	Mixed Waste	4	Tonnes		Tonnes	HARU HARU
Week 4	Batteries		Numbers		Numbers	
	E-Waste		Kilogram		Kilogram	
	Oil (Kitchen)	540	Liters		Liters	
	Oil (Engine)	1000	Liters		Liters	
	Food Waste		Kg		Kg	
	Mixed Waste	4	Tonnes		Tonnes	HARU HARU
Week 5	Batteries		Numbers		Numbers	
	E-Waste		Kilogram		Kilogram	
	Oil (Kitchen)		Liters		Liters	
	Oil (Engine)		Liters		Liters	
	Food Waste		Kg		Kg	
	Mixed Waste		Tonnes		Tonnes	

Total for the Month:	Total number of Batteries:	
	Total KG of E-Waste:	
	Total Liters of Oil Kitchen:	540
	Total Liters of Oil engine:	1000
	Total Tonne of Mixed Waste:	32

Checked by	Name: Faisal Ibrahim
Date:	30-11-21
Signature:	

Figure C 2: Waste generated for November 2021

October 2021

Waste Sent to Land Fills - Track Sheet - FS LG							
Month: Oct 2021							
Week	Waste Material	Boat Trip 1		Boat Trip 2		Name	Sign
Week 1	Batteries		Numbers		Numbers		
	E-Waste		Kilogram		Kilogram		
	Oil (Kitchen)		Liters		Liters		
	Oil (Engine)		Liters		Liters		
	Food Waste		Kg		Kg		
	Mixed Waste	16	Tonnes	4	Tonnes	HARM	HARM
Week 2	Batteries		Numbers		Numbers		
	E-Waste		Kilogram		Kilogram		
	Oil (Kitchen)		Liters		Liters		
	Oil (Engine)		Liters		Liters		
	Food Waste		Kg		Kg		
	Mixed Waste	4	Tonnes	16	Tonnes	HARM	HARM
Week 3	Batteries		Numbers		Numbers		
	E-Waste		Kilogram		Kilogram		
	Oil (Kitchen)		Liters		Liters		
	Oil (Engine)		Liters		Liters		
	Food Waste		Kg		Kg		
	Mixed Waste	4	Tonnes	4	Tonnes	HARM	HARM
Week 4	Batteries		Numbers		Numbers		
	E-Waste		Kilogram		Kilogram		
	Oil (Kitchen)		Liters		Liters		
	Oil (Engine)		Liters		Liters		
	Food Waste		Kg		Kg		
	Mixed Waste	16	Tonnes	4	Tonnes	HARM	HARM
Week 5	Batteries		Numbers		Numbers		
	E-Waste		Kilogram		Kilogram		
	Oil (Kitchen)		Liters		Liters		
	Oil (Engine)		Liters		Liters		
	Food Waste		Kg		Kg		
	Mixed Waste	4	Tonnes	4	Tonnes	HARM	HARM

Total for the Month:	Total number of Batteries:	
	Total KG of E-Waste:	
	Total Liters of Oil Kitchen:	
	Total Liters of Oil engine:	
	Total Tonne of Mixed Waste:	

Checked by
Name:
Date:
Signature:

Figure C 3: Waste generated for October 2021

September 2021

Waste Sent to Land Fills - Track Sheet - FS LG							
Month: Sep 2021							
Week	Waste Material	Boat Trip 1		Boat Trip 2		Name	Sign
Week 1	Batteries		Numbers		Numbers		
	E-Waste		Kilogram		Kilogram		
	Oil (Kitchen)		Liters		Liters		
	Oil (Engine)		Liters		Liters		
	Food Waste		Kg		Kg		
	Mixed Waste	16	Tonnes	4	Tonnes	Nabir	<i>[Signature]</i>
Week 2	Batteries		Numbers		Numbers		
	E-Waste		Kilogram		Kilogram		
	Oil (Kitchen)		Liters		Liters		
	Oil (Engine)		Liters		Liters		
	Food Waste		Kg		Kg		
	Mixed Waste	4	Tonnes	16	Tonnes	Nabir	<i>[Signature]</i>
Week 3	Batteries		Numbers		Numbers		
	E-Waste		Kilogram		Kilogram		
	Oil (Kitchen)		Liters		Liters		
	Oil (Engine)		Liters		Liters		
	Food Waste		Kg		Kg		
	Mixed Waste	4	Tonnes	4	Tonnes	HAKIR	HAKIR
Week 4	Batteries		Numbers		Numbers		
	E-Waste		Kilogram		Kilogram		
	Oil (Kitchen)		Liters		Liters		
	Oil (Engine)		Liters		Liters		
	Food Waste		Kg		Kg		
	Mixed Waste	4	Tonnes	4	Tonnes	HAKIR	HAKIR
Week 5	Batteries		Numbers		Numbers		
	E-Waste		Kilogram		Kilogram		
	Oil (Kitchen)	540	Liters		Liters		
	Oil (Engine)	1000	Liters		Liters		
	Food Waste		Kg		Kg		
	Mixed Waste	4	Tonnes	4	Tonnes	HAKIR	HAKIR

Total for the Month:	Total number of Batteries:	
	Total KG of E-Waste:	
	Total Liters of Oil Kitchen:	540 Liters
	Total Liters of Oil engine:	1000 Liters
	Total Tonne of Mixed Waste:	64

Checked by	
Name:	Faisal Dhorah
Date:	20.09.22
Signature:	<i>[Signature]</i>

Figure C 4: Waste generated for September 2021

August 2021

Waste Sent to Land Fills - Track Sheet - FS LG						
Month: Aug 2021						
Week	Waste Material	Boat Trip 1		Boat Trip 2		Name
Week 1	Batteries		Numbers		Numbers	
	E-Waste		Kilogram		Kilogram	
	Oil (Kitchen)		Liters		Liters	
	Oil (Engine)		Liters		Liters	
	Food Waste		Kg		Kg	
	Mixed Waste	16.3	Tonnes	3.5	Tonnes	Nabir
Week 2	Batteries		Numbers		Numbers	
	E-Waste		Kilogram		Kilogram	
	Oil (Kitchen)		Liters		Liters	
	Oil (Engine)		Liters		Liters	
	Food Waste		Kg		Kg	
	Mixed Waste	4.3	Tonnes	4	Tonnes	Nabir
Week 3	Batteries		Numbers		Numbers	
	E-Waste		Kilogram		Kilogram	
	Oil (Kitchen)		Liters		Liters	
	Oil (Engine)		Liters		Liters	
	Food Waste		Kg		Kg	
	Mixed Waste	16.3	Tonnes	4	Tonnes	Nabir
Week 4	Batteries		Numbers		Numbers	
	E-Waste		Kilogram		Kilogram	
	Oil (Kitchen)		Liters		Liters	
	Oil (Engine)		Liters		Liters	
	Food Waste		Kg		Kg	
	Mixed Waste	4	Tonnes	4	Tonnes	Nabir
Week 5	Batteries		Numbers		Numbers	
	E-Waste		Kilogram		Kilogram	
	Oil (Kitchen)	540	Liters		Liters	
	Oil (Engine)	1000	Liters		Liters	
	Food Waste		Kg		Kg	
	Mixed Waste	4	Tonnes	4	Tonnes	Nabir

Total for the Month:	Total number of Batteries:	
	Total KG of E-Waste:	
	Total Liters of Oil Kitchen:	540
	Total Liters of Oil engine:	1000
	Total Tonne of Mixed Waste:	60.3 tonnes

Checked by	
Name:	Faisal
Date:	30.08.21
Signature:	

Figure C 5: Waste generated for August 2021

July 2021

Waste Sent to Land Fills - Track Sheet - FS LG						
Month: July 2021						
Week	Waste Material	Boat Trip 1		Boat Trip 2		Name
Week 1	Batteries		Numbers		Numbers	
	E-Waste		Kilogram		Kilogram	
	Oil (Kitchen)		Liters		Liters	
	Oil (Engine)		Liters		Liters	
	Food Waste		Kg		Kg	
	Mixed Waste	4.5	Tonnes	4.5	Tonnes	Nabir
Week 2	Batteries		Numbers		Numbers	
	E-Waste		Kilogram		Kilogram	
	Oil (Kitchen)		Liters		Liters	
	Oil (Engine)		Liters		Liters	
	Food Waste		Kg		Kg	
	Mixed Waste	4	Tonnes	4	Tonnes	Nabir
Week 3	Batteries		Numbers		Numbers	
	E-Waste		Kilogram		Kilogram	
	Oil (Kitchen)		Liters		Liters	
	Oil (Engine)		Liters		Liters	
	Food Waste		Kg		Kg	
	Mixed Waste	168	Tonnes	3	Tonnes	Nabir
Week 4	Batteries		Numbers		Numbers	
	E-Waste		Kilogram		Kilogram	
	Oil (Kitchen)		Liters		Liters	
	Oil (Engine)		Liters		Liters	
	Food Waste		Kg		Kg	
	Mixed Waste	4	Tonnes	4	Tonnes	Nabir
Week 5	Batteries		Numbers		Numbers	
	E-Waste		Kilogram		Kilogram	
	Oil (Kitchen)	540	Liters		Liters	
	Oil (Engine)	1000	Liters		Liters	
	Food Waste		Kg		Kg	
	Mixed Waste	4	Tonnes	5	Tonnes	Nabir

Total for the Month:	Total number of Batteries:	
	Total KG of E-Waste:	
	Total Liters of Oil Kitchen:	540
	Total Liters of Oil engine:	1000
	Total Tonne of Mixed Waste:	49

Checked by
 Name: Faissal Dbrah'm
 Date: 31.07.21
 Signature: *[Signature]*

Figure C 6: Waste generated for July 2021

June 2021

Waste Sent to Land Fills - Track Sheet - FS LG						
Month: June 2021						
Week	Waste Material	Boat Trip 1	Boat Trip 2	Name	Sign	
Week 1	Batteries		Numbers			
	E-Waste		Kilogram			
	Oil (Kitchen)		Liters			
	Oil (Engine)		Liters			
	Food Waste		Kg			
	Mixed Waste	16	Tonnes	4	Nabir	AB
Week 2	Batteries		Numbers			
	E-Waste		Kilogram			
	Oil (Kitchen)		Liters			
	Oil (Engine)		Liters			
	Food Waste		Kg			
	Mixed Waste	4	Tonnes	16	Nabir	AB
Week 3	Batteries		Numbers			
	E-Waste		Kilogram			
	Oil (Kitchen)		Liters			
	Oil (Engine)		Liters			
	Food Waste		Kg			
	Mixed Waste	4	Tonnes	4	Nabir	AB
Week 4	Batteries		Numbers			
	E-Waste		Kilogram			
	Oil (Kitchen)		Liters			
	Oil (Engine)		Liters			
	Food Waste		Kg			
	Mixed Waste	16	Tonnes	4	Nabir	AB
Week 5	Batteries		Numbers			
	E-Waste		Kilogram			
	Oil (Kitchen)	540	Liters			
	Oil (Engine)	800	Liters			
	Food Waste		Kg			
	Mixed Waste	4	Tonnes	4	Nabir	AB

Total for the Month:	Total number of Batteries:	
	Total KG of E-Waste:	
	Total Liters of Oil Kitchen:	540
	Total Liters of Oil engine:	800
	Total Tonne of Mixed Waste:	76 ton

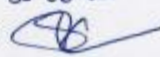
Checked by	
Name:	Faisal Dbrahim
Date:	30.06.21
Signature:	

Figure C 7: Waste generated for June 2021

May 2021

Waste Sent to Land Fills - Track Sheet - FS LG						
Month: May 2021						
Week	Waste Material	Boat Trip 1	Boat Trip 2	Name	Sign	
Week 1	Batteries	Numbers	Numbers			
	E-Waste	Kilogram	Kilogram			
	Oil (Kitchen)	Liters	Liters			
	Oil (Engine)	Liters	Liters			
	Food Waste	Kg	Kg			
	Mixed Waste	16 (Big Boat) Tonnes	4 Tonnes	Nabir	TS	
Week 2	Batteries	Numbers	Numbers			
	E-Waste	Kilogram	Kilogram			
	Oil (Kitchen)	Liters	Liters			
	Oil (Engine)	Liters	Liters			
	Food Waste	Kg	Kg			
	Mixed Waste	16 (Big Boat) Tonnes	4 Tonnes	Nabir	TS	
Week 3	Batteries	6 Batteries	Numbers			
	E-Waste	Kilogram	Kilogram			
	Oil (Kitchen)	Liters	Liters			
	Oil (Engine)	Liters	Liters			
	Food Waste	Kg	Kg			
	Mixed Waste	4 Tonnes	4 Tonnes	Nabir	TS	
Week 4	Batteries	Numbers	Numbers			
	E-Waste	Kilogram	Kilogram			
	Oil (Kitchen)	Liters	Liters			
	Oil (Engine)	Liters	Liters			
	Food Waste	Kg	Kg			
	Mixed Waste	16 (Big Boat) Tonnes	4 Tonnes	Nabir	TS	
Week 5	Batteries	Numbers	Numbers			
	E-Waste	Kilogram	Kilogram			
	Oil (Kitchen)	540 Liters	Liters	Nabir	TS	
	Oil (Engine)	1000 Liters	Liters			
	Food Waste	Kg	Kg			
	Mixed Waste	14 Tonnes	5 Tonnes	Nabir	TS	
Total for the Month:	Total number of Batteries:	6 batteries		Checked by		
	Total KG of E-Waste:			Name: Faizal		
	Total Liters of Oil Kitchen:	540 Liters		Date: 31.05.21		
	Total Liters of Oil engine:	1000 Liters		Signature: 		
	Total Tonne of Mixed Waste:	79 ton				

Figure C 8: Waste generated for May 2021

April 2021

Waste Sent to Land Fills - Track Sheet - FS LG							
Month: April 2021							
Week	Waste Material	Boat Trip 1		Boat Trip 2		Name	Sign
Week 1	Batteries		Numbers		Numbers		
	E-Waste		Kilogram		Kilogram		
	Oil (Kitchen)		Liters		Liters		
	Oil (Engine)		Liters		Liters		
	Mixed Waste	45	Tonnes	4	Tonnes	Nabir	AFW
Week 2	Batteries		Numbers		Numbers		
	E-Waste		Kilogram		Kilogram		
	Oil (Kitchen)		Liters		Liters		
	Oil (Engine)		Liters		Liters		
	Mixed Waste	4	Tonnes	4	Tonnes	Nabir	AFW
Week 3	Batteries		Numbers		Numbers		
	E-Waste		Kilogram		Kilogram		
	Oil (Kitchen)		Liters		Liters		
	Oil (Engine)		Liters		Liters		
	Mixed Waste	✓ 24	Tonnes	7.4	Tonnes	Nabir	AFW
Week 4	Batteries		Numbers		Numbers		
	E-Waste		Kilogram		Kilogram		
	Oil (Kitchen)		Liters	520	Liters		
	Oil (Engine)		Liters	225	Liters		
	Mixed Waste	35	Tonnes	7.00	Tonnes	Nabir	AFW
Week 5	Batteries	35	Numbers		Numbers		
	E-Waste		Kilogram		Kilogram		
	Oil (Kitchen)		Liters	360	Liters		
	Oil (Engine)		Liters		Liters		
	Mixed Waste	5	Tonnes	7	Tonnes	Nabir	AFW

Total for the Month:	Total number of Batteries:	
	Total KG of E-Waste:	
	Total Liters of Oil Kitchen:	520 Ltr.
	Total Liters of Oil engine:	225 Ltr.
	Total Tonne of Mixed Waste:	50 ton

Checked by	
Name:	Faisal Ibrahim
Date:	30.04.21
Signature:	

Figure C 9: Waste generated for April 2021

March 2021

Waste Sent to Land Fills - Track Sheet - FS LG							
Month: March 2021							
Week	Waste Material	Boat Trip 1	Boat Trip 2	Name	Sign		
Week 1	Batteries		Numbers				
	E-Waste		Kilogram				
	Oil (Kitchen)		Liters				
	Oil (Engine)		Liters				
	Mixed Waste	✓ 4	Tonnes	5	Tonnes	Nabib	DPH
Week 2	Batteries		Numbers				
	E-Waste		Kilogram				
	Oil (Kitchen)		Liters				
	Oil (Engine)		Liters				
	Mixed Waste	450	Tonnes	5	Tonnes	Nabib	DPH
Week 3	Batteries		Numbers				
	E-Waste		Kilogram				
	Oil (Kitchen)		Liters				
	Oil (Engine)		Liters				
	Mixed Waste	✓ 4	Tonnes	12/13	Tonnes	Nabib	DPH
Week 4	Batteries		Numbers				
	E-Waste		Kilogram				
	Oil (Kitchen)		Liters				
	Oil (Engine)		Liters				
	Mixed Waste	4	Tonnes	4	Tonnes		
Week 5	Batteries		Numbers				
	E-Waste		Kilogram				
	Oil (Kitchen)	630	Liters	35 can	Liters		
	Oil (Engine)	1000	Liters	5 Baral	Liters		
	Mixed Waste	7	Tonnes	5	Tonnes	Nabib	DPH
Total for the Month:	Total number of Batteries:						
	Total KG of E-Waste:						
	Total Liters of Oil Kitchen:		630 Liters				
	Total Liters of Oil engine:		1000 Liters				
	Total Tonne of Mixed Waste:		64 tonnes				

Checked by
 Name: faisal Ibrahim
 Date: 30.03.21
 Signature: *[Signature]*

Figure C 10: Waste generated for March 2021

February 2021

Waste Sent to Land Fills - Track Sheet - FS LG						
Month: February 2021						
Week	Waste Material	Boat Trip 1		Boat Trip 2		Name
			Numbers		Numbers	Sign
Week 1	Batteries					
	E-Waste		Kilogram		Kilogram	
	Oil (Kitchen)		Liters		Liters	
	Oil (Engine)		Liters		Liters	
	Mixed Waste	8.5	Tonnes	8.5	Tonnes	HARUN HARUN
Week 2	Batteries	04	Numbers		Numbers	
	E-Waste		Kilogram		Kilogram	
	Oil (Kitchen)		Liters		Liters	
	Oil (Engine)		Liters		Liters	
	Mixed Waste	A	Tonnes	8.5	Tonnes	HARUN HARUN
Week 3	Batteries		Numbers		Numbers	
	E-Waste		Kilogram		Kilogram	
	Oil (Kitchen)		Liters		Liters	
	Oil (Engine)		Liters		Liters	
	Mixed Waste	4+16	Tonnes	4	Tonnes	HARUN HARUN
Week 4	Batteries		Numbers		Numbers	
	E-Waste		Kilogram		Kilogram	
	Oil (Kitchen)	592	Liters		Liters	
	Oil (Engine)	1000	Liters		Liters	
	Mixed Waste	5	Tonnes	2.5	Tonnes	HARUN HARUN
Week 5	Batteries		Numbers		Numbers	
	E-Waste		Kilogram		Kilogram	
	Oil (Kitchen)		Liters		Liters	
	Oil (Engine)		Liters		Liters	
	Mixed Waste		Tonnes		Tonnes	

Total for the Month:	Total number of Batteries:	4
	Total KG of E-Waste:	
	Total Liters of Oil Kitchen:	592
	Total Liters of Oil engine:	1000
	Total Tonne of Mixed Waste:	48.5

Checked by
 Name: FAISAL IBRAHIM
 Date: 28th Feb 2021
 Signature: 

Figure C 11: Waste generated for February 2021

January 2021

Waste Sent to Land Fills - Track Sheet - FS LG							
Month: January 2021							
Week	Waste Material	Boat Trip 1		Boat Trip 2		Name	Sign
Week 1	Batteries		Numbers		Numbers		
	E-Waste		Kilogram		Kilogram		
	Oil (Kitchen)		Liters		Liters		
	Oil (Engine)		Liters		Liters		
	Mixed Waste	4	Tonnes	3.5	Tonnes	Haram	HARAM
Week 2	Batteries		Numbers		Numbers		
	E-Waste		Kilogram		Kilogram		
	Oil (Kitchen)		Liters		Liters		
	Oil (Engine)		Liters		Liters		
	Mixed Waste	4	Tonnes	4	Tonnes	Haram	HARAM
Week 3	Batteries		Numbers		Numbers		
	E-Waste		Kilogram		Kilogram		
	Oil (Kitchen)		Liters		Liters		
	Oil (Engine)		Liters		Liters		
	Mixed Waste	4	Tonnes	4	Tonnes	Haram	HARAM
Week 4	Batteries		Numbers		Numbers		
	E-Waste		Kilogram		Kilogram		
	Oil (Kitchen)	720	Liters		Liters		
	Oil (Engine)	1000	Liters		Liters		
	Mixed Waste	3	Tonnes	4.5	Tonnes	Haram	HARAM
Week 5	Batteries		Numbers		Numbers		
	E-Waste		Kilogram		Kilogram		
	Oil (Kitchen)		Liters		Liters		
	Oil (Engine)		Liters		Liters		
	Mixed Waste		Tonnes		Tonnes		
Total for the Month:	Total number of Batteries:						
	Total KG of E-Waste:						
	Total Liters of Oil Kitchen:	720 Liter					
	Total Liters of Oil engine:	1000 Liter					
	Total Tonne of Mixed Waste:	31 tonnes					

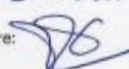
Checked by
Name: FAISAL IBRAHIM
Date: 31.01.21
Signature: 

Figure C 12: Waste generated for January 2021

B. Garbage boat hire and Thillafushi gate fee payment

Four Seasons Landaa Giraavaru							
Record of Garbage Boat hire 2021							
PO number	Received date	Product name	UOM	Number of trips	Estimated capacity of each trip (Tonnage)	Total Tonnage	Unit price (MVR)
MLG0030453	16/02/2021	Garbage dhoni monthly rent charge	EA	4	8.5	34	62,000.00
MLG0031727	23/03/2021	Garbage dhoni monthly rent charge	EA	4	8.5	34	62,000.00
MLG0031536	21/04/2021	Garbage dhoni monthly rent charge	EA	4	8.5	34	62,000.00
MLG0031788	15/05/2021	Garbage dhoni monthly rent charge	EA	4	8.5	34	62,000.00
MLG0034764	13/06/2021	Garbage dhoni monthly rent charge	EA	4	8.5	34	62,000.00
MLG0041835	16/11/2021	Garbage dhoni monthly rent charge	EA	4	8.5	34	62,000.00
MLG0030817	13/08/2021	Garbage dhoni monthly rent charge	EA	4	8.5	34	62,000.00
MLG0038234	12/08/2021	Garbage dhoni monthly rent charge	EA	4	8.5	34	62,000.00
MLG0039003	18/10/2021	Garbage dhoni monthly rent charge	EA	4	8.5	34	62,000.00
MLG0040357	19/11/2021	Garbage dhoni monthly rent charge	EA	4	8.5	34	62,000.00
MLG0038213	11/07/2021	Garbage dhoni monthly rent charge	EA	4	8.5	34	62,000.00
MLG0029013	04/01/2021	Garbage Dhoni Hire Charge	EA	3	16.5	49.5	20,000.00
MLG0029461	18/01/2021	Garbage Dhoni Hire Charge	EA	3	16.5	49.5	40,000.00
MLG0030465	14/02/2021	Garbage Dhoni Hire Charge	EA	3	16.5	49.5	40,000.00
MLG0031344		Garbage Dhoni Hire Charge	EA	2	16.5	33	20,000.00
MLG0031529		Garbage Dhoni Hire Charge	EA	2	16.5	33	20,000.00
MLG0031891	19/03/2021	Garbage Dhoni Hire Charge	EA	3	16.5	49.5	20,000.00
MLG0033393	27/04/2021	Garbage Dhoni Hire Charge	EA	3	16.5	49.5	20,000.00
MLG0032158	25/05/2021	Garbage Dhoni Hire Charge	EA	3	16.5	49.5	20,000.00
MLG0032607		Garbage Dhoni Hire Charge	EA	3	16.5	49.5	20,000.00
MLG0033539	02/05/2021	Garbage Dhoni Hire Charge (Hiring Date - 29th Apr 21 & 0	EA	2	16.5	33	20,000.00
MLG0034825	10/06/2021	Garbage Dhoni Hire Charge (Hiring Date 03rd & 09th June	EA	3	16.5	49.5	40,000.00
MLG0034558	05/06/2021	Garbage Dhoni Hire Charge (Hiring Date 09th May 21)	EA	3	16.5	49.5	40,000.00
MLG0031776	09/01/2021	Garbage Dhoni Hire Charge (Hiring Date 09th May 21)	EA	3	16.5	49.5	20,000.00
MLG0031568	20/06/2021	Garbage Dhoni Hire Charge (Hiring Date 12th & 20th June)	EA	3	16.5	49.5	40,000.00
MLG0037851	14/08/2021	Garbage Dhoni Hire Charge (Hiring Date 28th July 21)	EA	3	16.5	49.5	20,000.00
MLG0037375	22/08/2021	Garbage Dhoni Hire Charge (Hiring Date 28th July 21)	EA	3	16.5	49.5	20,000.00
MLG0039664	27/10/2021	Garbage Dhoni Hire Charge (Hiring Date 28th July 21)	EA	3	16.5	49.5	20,000.00
MLG0037976	08/09/2021	Garbage Dhoni Hire Charge (Hiring Date 28th July 21)	EA	3	16.5	49.5	20,000.00
MLG0038647	02/10/2021	Garbage Dhoni Hire Charge (Hiring Date 28th July 21)	EA	3	16.5	49.5	20,000.00
MLG0038900	10/10/2021	Garbage Dhoni Hire Charge (Hiring Date 28th July 21)	EA	3	16.5	49.5	20,000.00
MLG0036508	29/07/2021	Garbage Dhoni Hire Charge (Hiring Date 28th July 21)	EA	3	16.5	49.5	20,000.00
MLG0037780	01/09/2021	Garbage Dhoni Hire Charge (Hiring Date 28th July 21)	EA	3	16.5	49.5	20,000.00
MLG0040956	13/12/2021	Garbage Dhoni Hire Charges from LG to Thilafushi	EA	3	16.5	49.5	15,000.00
MLG0040957	13/12/2021	Garbage Dhoni Hire Charges from LG to Thilafushi	EA	3	16.5	49.5	20,000.00
MLG0041703	26/12/2021	Transportation of garbage from LG to Thilafushi	EA	3	16.5	49.5	20,000.00
MLG0041410	15/12/2021	Transportation of garbage from LG to Thilafushi	EA	3	16.5	49.5	20,000.00
Totals				73 Boat trips		853 Tonne	
Prepared by:			Information received from:				
Name:	Faisal Ibrahim		Name:	Ganpat Dalvi			
Designation:	Environment and Sustainability Leader		Designation:	Director of Purchasing			
Date:	12/01/2022		Date:	12/01/2022			

Figure C 13: Four Seasons Landaa Giraavaru records of payments to WAMCO and garbage dhoni

C. Waste disposal cost

AY **Abdul Nasir Yousuf**
Tue 3/22/2022 6:07 AM

To: MLG EngCoordinator
Cc: Ganpat Dalvi; Prabhu Neelakandan

Dear Faisal,

Please see requested details below,

Garbage Dhoni USD4020.75/Month
WAMCO USD2284.82/Month

This charges will be same always until otherwise we didn't hire extra garbage dhoni.

Extra Garbage dhoni rent USD1297.02/Trip
WAMCO Depending on tonnage of dhoni.

Figure C 14: Waste disposal costs as per March 2022

D. Secure Bag Pte Ltd price list



Sl No.	We are pleased to quote our best rates as follows.		Rate (in MVR)
	Description		To Thilafushi / Male
1	Iron Scrap (light Metal) such as can, roofing sheet, Cylinders etc,		1.50
2	Iron scrap heavy metal		2.50
3	Batteries		7.00
4	motors/ Stainless steel		6.00
5	Glass (in powdered form)		0.25
6	Copper (cleaned)		85.00
7	Copper Wire (un cleaned)		20.00
8	Brass scraps		35.00
9	Aluminium scraps		8.00
10	Can Aluminium (bundled)		5.00
11	Air conditioners		6.00
12	Used Lube Oil (per Drum 200 ltr)		45.00
13	Used cooking oil (per drum 200 ltr)		45.00

Figure C 15: Secure bag Pte Ltd price list for recyclables